A.T.L.A.S. (AUTOMATED TIMETABLING AND LOCATIONS ALLOCATION SYSTEM): A WEB-BASED ACADEMIC SCHEDULING APPLICATION

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INTRODUCTION

Background of the Study

Educational institutions globally are digitizing timetabling to address the critical errors inherent in manual scheduling. Traditional methods frequently lead to double-bookings and misallocated resources (Nsulangi, P. T., et. al., 2024). This shift toward automation is evident in institutions such as Indonesia's SMPN 1 Jombang (Prakasa, T. A. D., et. al., 2024). By reducing calculation errors and adapting to dynamic changes (Nsulangi, P. T., et. al., 2024), automated systems offer a sustainable solution to administrative bottlenecks worldwide (Shomoye, M. A. & Mosaku, A. O., 2025).

Despite this global shift, several Philippine institutions still rely on partially manual methodologies. Recent local studies highlight approaches utilizing spreadsheets, as seen in the timetabling models of the University of Baguio (Tayong, A. A., & Roma, G. M., 2023) and the City College of Angeles (Yee, B. E. D., et. al., 2024). Although systematized timetabling optimizes resource allocation (Romaguera, D., et. al., 2024), spreadsheet tools lack the capacity to independently optimize schedules. They successfully store and computerize data but still require tedious manual input to generate new timetables.

These national administrative challenges directly impact local public Junior High Schools (JHS), specifically Hinigaran National High School. Preliminary discussions with personnel from the pilot site reveal significant issues with rigid setups that struggle to adapt to varying institutional needs. Effective class allocation requires balancing physical room capacities alongside critical soft constraints. Unfortunately, manual methods often ignore important factors like teacher travel time and teacher workload. These core issues are further compounded by a lack of real-time updates for sudden schedule adjustments and unstable local internet infrastructure that renders purely cloud reliant solutions unreliable. Therefore, there is a clear need for a configurable, centralized algorithm tailored for the Junior High School level that addresses these specific physical, human, and infrastructure limitations.

To address these interconnected gaps, this study proposes A.T.L.A.S. (Automated Timetabling and Locations Allocation System). This mobile accessible web application is specifically designed to balance competing scheduling goals within Junior High Schools. Deploying the software at the pilot site will directly evaluate its configurable architecture. A.T.L.A.S. features a priority management module for administrators to dynamically adjust rules, effectively balancing specialized room matching against teacher workload limits. The core engine then generates conflict-free timetables based on the submitted data and configured constraints, while a visual drag-and-drop interface allows for seamless manual review and fine-tuning. Finalized schedules are then deployed to a real-time, synchronized dashboard for teacher and students, bridging the communication gap. Finally, the software will undergo a rigorous ISO 25010 evaluation to ensure reliability, performance, and stakeholder trust.

Objective of the Study

The general objective of the study aims to design and develop a mobile-responsive web application titled "A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application".

Specifically, this study aims:

1. to design and develop a web and mobile application with the following technical features:

    1.1. web-based administrative portal with drag-and-drop scheduling interface.

    1.2. configurable scheduling priority management module.

    1.3. mobile teacher authentication with schedule preference submission.

    1.4. automated timetable generation system.

2. to test the functionality of the features mentioned above

3. to evaluate the system using ISO 25010 standardize evaluation tool in terms of Functionality Suitability, Performance Efficiency, Usability, Compatibility, Reliability, Security, Maintainability, and Portability.

Significance of the Study

The A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application is designed to be significant and beneficial to the following groups:

The School Scheduler. This study provides a practical, automated solution that directly resolves the administrative bottlenecks of manual timetabling. By replacing spreadsheet data entry with an algorithmic engine and a visual drag-and-drop interface, the research demonstrates how automated priority management can instantly generate conflict-free schedules, saving schedulers weeks of productivity.

The Teachers. The implementation of this system proves the critical importance of integrating human-centric "soft constraints" into educational technology. Instead of randomized schedules, the system actively minimizes physical travel time between distant classrooms and prevents consecutive, heavy teaching loads, providing a working model of how digital tools can optimize occupational efficiency and ensure instructional sustainability by mitigating scheduling fatigue.

The Students. Students benefit directly from the system's real-time, synchronized, mobile-friendly dashboard. The study demonstrates how eliminating uncommunicated, last-minute room changes fosters a more organized academic environment, ensuring that students are consistently and accurately allocated to appropriately sized, specialized learning facilities.

The Schools. For the participating pilot institution, Hinigaran National High School, this research serves as a strategic blueprint for institutional digital transformation. It offers a tested framework showing how implementing "hard constraints" in a centralized algorithm guarantees that physical facilities, specialized laboratories, and room capacities are maximized efficiently without double-bookings or wasted space.

Researchers. This study provides the researchers with a practical opportunity to apply their theoretical knowledge in software engineering, database management, and algorithmic logic. By accomplishing this study and developing a real-world system, this allows researcher to enhance their technical proficiency and positively challenges their technical skills.

Future Researchers. This study serves as a valuable literature reference for future researchers exploring multi-objective optimization algorithms. By detailing a generic, highly configurable architecture and subjecting it to the rigorous ISO 25010 software quality framework, this research provides a foundational academic baseline for expanding automated scheduling solutions into Senior High School or Tertiary educational levels.

Scope and Limitation of the Study

This study covers the design, development, and evaluation of A.T.L.A.S. (Automated Timetabling and Locations Allocation System), a mobile-responsive web application designed to automate academic scheduling for Junior High Schools. Pilot testing will be conducted exclusively at Hinigaran National High School. The system features a web-based administrative dashboard where schedulers can set specific priorities such as room capacities, laboratory requirements, and teacher mobility constraints and spatial preferences, such as restricting room assignments to the ground floor for pregnant personnels or teachers with disabilitites, and use a visual, interactive grid with drag-and-drop capabilities to manually fine-tune room allocations. To ensure the schedule is accurate, teachers are provided with a mobile-friendly portal to directly submit their availability and health preferences before the algorithm runs. Using this collected data, the system automatically generates a conflict-free timetable to prevent double-bookings.

Finally, students can access a synchronized, mobile-responsive platform to view real-time updates of their finalized schedules and easily download them as images.

Despite its automated capabilities, the system operates strictly within specific boundaries and is not intended to replace the human administrator. Rather, it serves as a supportive tool that improves efficiency, reliability, and productivity in schedule planning. Recognizing that stable internet access cannot always be guaranteed in school environments, A.T.L.A.S. is designed to be offline-capable for selected functions, rather than fully offline for all operations. The system remains accessible through a web browser without requiring mobile app installation, and core interface resources can still load on previously used devices even when connectivity is limited. In addition, certain user actions completed during a connection interruption, such as preference entries or schedule adjustments, may be temporarily stored and synchronized once access to the main server is restored. Previously viewed schedule pages may also remain available for reference. However, timetable generation, publication of finalized schedules, and distribution of updated data across all users still depend on connection to the central server where authoritative school records are maintained. In practical terms, this means that users connected to the same local school network can continue operating even without public internet, while users outside that network require internet-enabled access. Functionally, the system's output quality still relies on complete and accurate data entry; if rules are conflicting or faculty inputs are incomplete, the algorithm's effectiveness is reduced and manual correction becomes necessary. Furthermore, the platform is limited to scheduling workflows and does not process enrollment, grades, or attendance, and it does not include student-side editing or communication features since the student portal is strictly view-only.

Definition of Terms

For a clearer understanding of the study, the following terms are defined both contextually based on existing literature and operationally as they are used in the context of A.T.L.A.S.

A.T.L.A.S. (Automated Timetabling and Locations Allocation System). Operationally, this is the mobile-responsive web application designed and developed in this study to automate the academic scheduling process for local Junior High Schools through algorithmic optimization and manual refinement.

Automation. Contextually, this refers to the use of intelligent algorithms and computerized systems to automatically construct schedules, resolve overlaps, and allocate resources with minimal human intervention (Shomoye & Mosaku, 2025).

Operationally, in this study, it refers to the system's ability to utilize a computational algorithm to independently generate conflict-free academic schedules, replacing the traditional manual data entry in spreadsheet software.

Background Data Synchronization. Contextually, it refers to an automated process in modern web architecture where an application temporarily stores user actions locally while offline and seamlessly uploads them to a central server without requiring manual user intervention once network connectivity is restored (Venzano Rodriguez, 2022).

Operationally, it is the capability of A.T.L.A.S. to safely save teacher preferences or administrative schedule adjustments locally on the device during an internet outage, and automatically upload that data to the main database the moment the connection returns.

Drag-and-Drop Interface. Contextually, this is a graphical user interface (GUI) interaction technique where users select a virtual object by "grabbing" it and moving it to a different location, facilitating intuitive and semi-automated data adjustments within web-based administrative systems (Garcia-Yañez et al., 2024).

Operationally, it is a visual, interactive dash Guia board feature that allows the school administrator to manually click, move, and seamlessly adjust computer-generated room allocations and time slots for fine-tuning purposes.

Genetic Algorithm. Contextually, this is an evolutionary search heuristic that mimics the process of natural selection to solve complex optimization problems through crossover and mutation. (Source: Katore et al., 2025).

Operationally, it serves as the core generation engine of the system that processes teacher preferences and resource constraints to output an optimal timetable.

Hard Constraints. Contextually, it is a rigid, essential conditions in educational timetabling that must be strictly satisfied for a schedule to be considered valid; violating them renders the solution unacceptable (Burke, Petrovic, & Qu, 2023).

Operationally, these are the absolute, non-negotiable physical rules the scheduling algorithm must satisfy to prevent logistical errors, such as strictly adhering to maximum room capacities, matching subjects to specialized laboratories, and preventing double-bookings of rooms or teachers.

ISO 25010. Contextually, it is the international software-engineering standard used to evaluate software quality attributes such as functional suitability, performance efficiency, and usability (Canlas, Piad, & Lagman, 2021)

Operationally, it is the standardized software quality framework used by the researchers to rigorously evaluate the final system's reliability, performance efficiency, and overall usability during the pilot testing phase.

Mobile-Responsive. Contextually, it is a modern web development architecture that dynamically adjusts a system's layout and functionalities depending on the device's screen size, ensuring seamless accessibility across smartphones and desktops without requiring native application installation (Venzano Rodriguez, 2022).

Operationally, it refers to the technical design architecture of the web application that allows it to automatically adapt and function optimally on smartphone screens for teachers and students, eliminating the need to download a native mobile application (APK).

Priority Management Module. Contextually, this represents a multi-objective weighting mechanism where administrators can assign mathematical significance to different scheduling goals, guiding the algorithm to prioritize certain soft constraints over others during generation (Oyelere et al., 2024).

Operationally, it is a configuration feature within the administrative portal that allows the scheduler to dynamically set and rank the importance of different scheduling rules, ensuring the algorithm aligns with the specific needs of the pilot school.

Rapid Application Development (RAD). Contextually, it is an agile, iterative SDLC methodology that prioritizes structural flexibility and rapid prototyping based on continuous user feedback (Source: Salesforce, n.d.).

Operationally, it is the development methodology used by the researchers to build, demonstrate, and refine the A.T.L.A.S. prototype approximately within 100 days timeframe.

Parameters. Contextually, these are the measurable factors, variables, or input data sets that define the operating conditions of a heuristic algorithm and guide its computational process to find an optimal scheduling solution (Chen et al., 2023).

Operationally, it represents the specific set of rules, physical room limits, and inputted teacher preferences that dictate the boundaries within which the algorithm must operate to generate a valid schedule.

Real-Time Synchronization. Contextually, it is the automated process of continuously coordinating and updating data across a distributed system simultaneously, ensuring immediate consistency across all viewing interfaces and preventing data silos (Yee et al., 2024).

Operationally, it is the capability of the system to instantly update the teacher and student view-only dashboards the exact moment an administrator finalizes a schedule or makes a manual room adjustment.

Soft Constraints. Contextually, these are the desirable, human-centric preferences that are not mandatory but improve schedule quality and acceptance when satisfied (Burke, Petrovic, & Qu, 2023).

Operationally, these are the highly desired, human-centric conditions the algorithm attempts to satisfy to protect teacher well-being, such as minimizing physical travel time between distant classrooms and preventing heavy, continuous teaching loads without breaks.

Timetabling. Contextually, it is defined as the allocation of resources to events placed in space-time to meet educational objectives while respecting physical and human constraints (Burke & Petrovic, 2023).

Operationally, it is the complex administrative process of plotting and allocating teachers, junior high school students, subjects, and physical classrooms into specific time slots without encountering logistical conflicts.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and studies from both local and foreign sources. The studies included in this chapter helps in familiarizing information that is relevant and similar to the present study.

Visual Interactive Interfaces and Drag-and-Drop Functionality in Scheduling

While algorithmic generation successfully resolves the heavy computational burden of academic scheduling, strict mathematical models often lack the nuanced flexibility required by human administrators. To bridge this gap, modern educational technology emphasizes pairing automatic engines with intuitive, visual interfaces. Research indicates that integrating drag-and-drop grids with color-coded alerts allows administrators to dynamically refine schedules, reducing overall scheduling time by up to 75% compared to conventional methods (Ghazal et al., 2025).

A distinct architectural and financial gap exists between generic commercial software and tailored academic deployments. Commercial solutions like SchoolTracs and Lantiv function as expensive, generalized cloud-based products requiring continuous subscriptions (SchoolTracs, 2025 & Lantiv, 2026). While they offer robust visual functionalities, they frequently present severe architectural and financial limitations for local institutions. Conversely, while open-source utilities like the FET Timetable Explorer remove the financial burden, they require complex desktop installations that lack the synchronized accessibility of live web portal (FET, 2025). In contrast, academic systems emphasize localized zero installation web deployments that ensure maximum administrative control without the burden of commercial dependencies or local installation hurdles.

A.T.L.A.S. integrates this proven interactive methodology to ensure maximum administrative control, but applies it within a locally adapted web architecture. Following the automated generation phase, the system provides a specialized drag-and-drop grid that empowers administrators to manually override allocations and fine-tune room assignments. This ensures the critical visual interface remains accessible, cost-effective, and precisely tailored to the target institution's operational constraints.

Configurable Priority Management and Human-Centric Soft Constraints

The transition from rigid algorithmic scheduling to adaptable, human-centric priority management is essential for institutional fairness and teacher well-being. Recent literature emphasizes that modern timetabling must extend beyond merely preventing spatial conflicts to strategically incorporate soft constraints, such as minimizing idle periods and balancing teaching loads. Studies evaluating complex high school scheduling models demonstrate that mathematically prioritizing these human parameters results in timetables that are significantly more practical and sustainable for educators (Sun & Wu, 2023). Furthermore, by formally defining pre-determined teaching load limits and teacher skills, administrative tools can generate objective assignments that successfully eliminate administrative bias and reduce workplace grievances (Khorbotly & White, 2024).

Within the educational context, the effectiveness of these automated systems relies heavily on how constraints are applied. A recent local evaluation highlighted that empowering administrator to dynamically configure scheduling criteria ensures the software successfully meets the specific institutional needs (Keh & Sarmiento, 2025).

A.T.L.A.S. directly applies this proven standard through its configurable priority management module. By providing a centralized, dynamic dashboard, the system allows school administrators to independently adjust the mathematical weights of various soft constraints, such as maximum teaching hours and department-level rules. This ensures the algorithmic engine consistently adapts to the school's unique policies while maintaining absolute systemic fairness and respecting teacher well-being.

Secure Teacher Portals and Decentralized Data Collection

Beyond secure authentication, the primary function of a dedicated teacher portal is the accurate, decentralized collection of scheduling parameters. Algorithmic timetabling models rely heavily on precise data inputs to maximize the alignment between course offerings and teacher capabilities. Recognizing the time of qualified instructors as a highly valuable academic resource, research argues that directly integrating teacher preferences into the scheduling engine is critical for optimal assignment (Khorbotly & White, 2024).

Traditional, centralized data entry often leads to administrative conflicts and wasted man-hours. However, recent technological deployments demonstrate that providing professors with web-based portals to autonomously enter their availability drastically reduces academic programming errors (Garcia Yañez et al., 2024). By decentralizing this data-entry process, institutions successfully eliminate the administrator as the middleman for data entry and prevent logistical bottlenecks.

A.T.L.A.S. directly addresses these operational necessities by incorporating a secure, mobile-responsive teacher portal linked seamlessly to its central timetable generator. This architecture empowers teacher members to safely and independently submit their teaching availability, preferred schedules, and soft constraints. Once submitted, the system automatically processes these variables during schedule generation to maximize resource efficiency and ensure the automated engine is fed accurate, real-time data directly from the instructor.

Automated Timetable Generation and Algorithmic Optimization

Educational timetabling is universally classified as an NP-hard problem, meaning the sheer volume of logistical constraints makes it mathematically infeasible to resolve manually without exhausting administrative resources (Tan et al., 2021, Yavuz et al., 2023). As secondary education curricula grow in complexity, modern decision-support systems must transition from manual plotting to advanced algorithmic optimization to manage escalating workloads.

Recent literature highlights varying success among computational approaches. While Sun and Wu (2023) and Tassopoulos et al. (2023) successfully applied two-phase tabu search and particle swarm optimization alongside the simulated annealing models of Prakasa et al. (2024) and Uysal et al. (2025) to manage standard teacher loading, these heuristic methods struggle to scale against severe spatial limitations. Conversely, international research proves that genetic algorithms excel at resolving dense bottlenecks through computational crossover and mutation where traditional models fail (Katore et al., 2024, Ranolo et al., 2025, Uma et al., 2024). Furthermore, local evaluations validate that enhanced genetic algorithms successfully optimize spatial resources and eliminate conflicts within the Philippine educational context (Romaguera et al., 2024).

To further optimize generation, recent studies employ hybrid architectures. Pairing a genetic algorithm with a greedy deterministic constructor for initial classroom allocation significantly reduces computational time and improves scheduling quality (Han & Wang, 2025, Fadilah et al., 2024). ATLAS integrates these proven principles into its core engine. By leveraging a greedy constructive foundation for rapid initial plotting followed by the adaptive mutation of a genetic algorithm, the system instantly outputs a mathematically feasible, conflict-free schedule. This tailored hybrid approach eliminates redundant calculations and compresses weeks of manual trial-and-error into a fraction of the execution time for Junior High Schools.

Table 1.

Matrix of Related Literatures and Studies

RRL Studies/System	Features web-based administrative portal with drag-and-drop scheduling interface.	configurable scheduling priority management module	mobile teacher authentication with schedule preference submission	. automated timetable generation system
1. Real-Time Modular Scheduling System for Educational Operations – Ghazal et al. (2025)	✓	✗	✗	✗
2. Timetabling Turbo 2026 [Computer software] – Lantiv (2026)	✓	✗	✗	✗
3. FET Timetable Explorer [Computer software] – FET (2025)	✓	✗	✗	✗
4. Easy drag-and-drop scheduling [Computer software] – SchoolTracs (2025)	✓	✗	✗	✗
5. A Preference-Based Faculty-Assignment Tool for Course Scheduling Optimization – Khorbotly & White (2024)	✗	✓	✓	✗
6. Automated Class Scheduling System For Aemilianum College Inc. – Keh & Sarmiento (2025)	✗	✓	✗	✗
7. Two-phase tabu search algorithm for solving Chinese high school timetabling problems under the new college entrance examination reform – Sun & Wu (2023)	✗	✓	✗	✓
8. A Preference-Based Faculty-Assignment Tool for Course Scheduling Optimization. 2024 ASEE Annual Conference & Exposition – Khorbotly & White (2024)	✗	✓	✓	✗
9. Development of a Web-Based Timetabling Software for a Mexican University – Garcia Yañez et al. (2024)	✗	✗	✓	✗
10. A Course Timetabling Problem for Classroom Usage Minimization – Yavuz et al. (2023)	✗	✗	✗	✓
11. A Web-based Decision Support System for Managing Course Timetabling in Online Education – Uysal et al. (2025)	✗	✗	✗	✓

12. A survey of the state-of-the-art of optimisation methodologies in school timetabling problems – Tan et al. (2021)	✗	✗	✗	✓
13. Two-phase tabu search algorithm for solving Chinese high school timetabling problems under the new college entrance examination reform – Sun & Wu (2023)	✗	✓	✗	✓
14. An Effective Local Particle Swarm Optimization-Based Algorithm for Solving the School Timetabling Problem – Tassopoulos et al. (2023)	✗	✗	✗	✓
15. Automating School Timetabling: An Intelligent System Application Using Simulated Annealing – Prakasa et al. (2024)	✗	✗	✗	✓
16. Dynamic Timetable Generator Using Genetic Algorithm – Katore et al. (2024-2025)	✗	✗	✗	✓
17. Automated timetable generation for academic institutions – Uma et al. (2024)	✗	✗	✗	✓
18. Development of a Web-based Course Timetabling System based on an Enhanced Genetic Algorithm – Romaguera et al. (2024)	✗	✗	✗	✓
19. A Hybrid Approach to Automatic Timetabling Using Self-Organizing Maps, Secure Convex Dominating Sets, and Metaheuristics – Ranolo et al. (2025)	✗	✗	✗	✓
20. A Web-based Decision Support System for Managing Course Timetabling in Online Education – Uysal et al. (2025)	✗	✗	✗	✓
21. Gradual Optimization of University Course Scheduling Problem Using Genetic Algorithm and Dynamic Programming – Han & Wang (2025)	✗	✗	✗	✓
22. A Comparison of Welch Powell Algorithm and Greedy Algorithm – Fadilah et al. (2024)	✗	✗	✗	✓

Synthesis

The review of related literature and systems establishes academic timetabling as a highly complex, NP-hard logistical challenge that consistently exceeds the capacity of manual, heuristic methodologies. To resolve the dense matrix of spatial limitations and human-centric variables, modern educational institutions necessitate advanced computational frameworks; consequently, A.T.L.A.S. employs Evolutionary and Genetic

Algorithms as its central generation engine to achieve mathematical optimization in resource allocation. However, current literature emphasizes that algorithmic efficacy is intrinsically dependent on accurate, decentralized data collection. The system fulfills this requirement by integrating secure teacher portals for direct availability input alongside a configurable priority management module, empowering administrators to mathematically weigh institutional soft constraints. Acknowledging that strict computational models cannot invariably anticipate real-world edge cases, A.T.L.A.S. bridges the divide between machine calculation and human oversight through an interactive drag-and-drop grid tailored for manual schedule refinement. Ultimately, the synthesis of these theoretical models and applied systems supports the architectural and design rationale of ATLAS. Rather than proposing a singular perfect solution, the literature confirms that integrating interactive visual interfaces, alongside mobile preference submission and a hybrid algorithmic engine, provides a highly appropriate framework for addressing the timetabling challenges of local Junior High Schools. By combining a configurable priority management module with a greedy deterministic constructor and genetic algorithm optimization, this research builds a context-specific solution tailored exactly to the operational needs of public education.

Chapter 3

DESIGN AND METHODOLOGY

Research Design

This study utilizes a combined Descriptive and Developmental Research Design to facilitate the creation, implementation, and evaluation of A.T.L.A.S. (Automated Timetabling and Locations Allocation System).

The Descriptive Research Design is employed to conduct a thorough qualitative needs assessment prior to system construction (Creswell & Creswell, 2018). As described by Purdy and Popan (n.d.), descriptive research allows for the detailed examination of situations and outcomes by prioritizing the subjects' authentic experiences without manipulating the research context. By analyzing the existing scheduling protocols of the pilot school, the researchers can identify critical parameters, such as teacher availability preferences, room capacity constraints, and institutional priorities. Understanding the what, why, and how of the current manual scheduling process provides the foundational logic required to accurately configure the automated timetable generation algorithm. Furthermore, the specific scheduling problems and limitations observed during the interviews will be translated into practical testing scenarios through the formulation of a Test Case Questionnaire. These scenarios are based on actual, past scheduling conflicts that the school has faced. The A.T.L.A.S. system must be able to successfully solve these real-world situations, which will serve as the primary standard to measure if the software works correctly.

Concurrently, the Developmental Research Design is integrated, as the primary objective of this study is the actual design, construction, and deployment of a functional software solution (Richey, 1994). The scope of developmental research explicitly encompasses the creation and evaluation of both instructional and non-instructional products and tools (Ibrahim, 2016). Therefore, the introduction of the A.T.L.A.S. system serves as a technological intervention—specifically, a non-instructional administrative tool—integrated into the current manual scheduling processes of the participating educational institution. To systematically execute this developmental process, the researchers will utilize the Rapid Application Development methodology as the primary software development life cycle. By employing an iterative prototype cycle of building, demonstrating, and refining, this methodology perfectly aligns with the developmental goal of creating a highly adaptable software intervention that directly responds to the qualitative needs gathered during the descriptive phase. By deploying the system and conducting rigorous testing alongside ISO 25010 standard evaluations, the researchers will track the resulting changes and improvements in the school's scheduling workflows. This approach allows the researchers to formally observe how the software intervention develops and enhances the efficiency, accuracy, and overall operational capacity of the system's users over the duration of the study.

System Development Life Cycle

For the design and engineering of A.T.L.A.S. (Automated Timetabling and Locations Allocation System), the researchers selected the Rapid Application Development

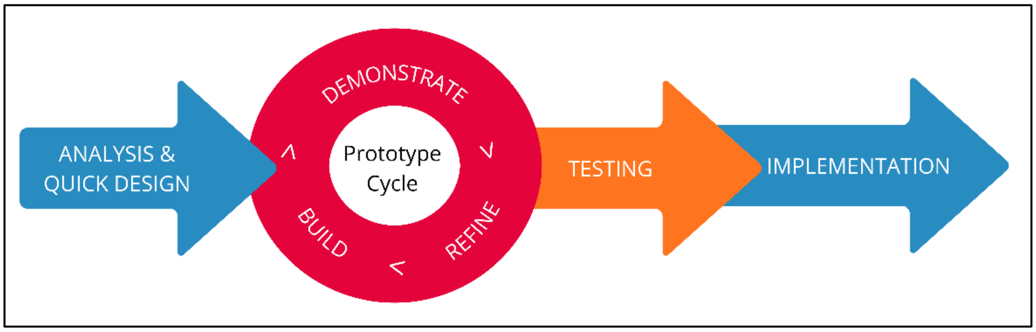
(RAD) model as the primary Software Development Life Cycle (SDLC) methodology. RAD supports an agile, iterative development process, allowing developers to quickly build, demonstrate, and refine prototypes based on direct user feedback. This flexibility ensures that specific system features and goals are reached without the need to restart the entire development lifecycle when requirements change (Salesforce, n.d.). Furthermore, RAD emphasizes a short, component-based software development cycle. This approach aligns significantly with the developmental objectives of A.T.L.A.S., enabling the successful delivery of full system functionality within a strict 60 to 90-day timeframe (Subhiyakto et al., 2019). While the A.T.L.A.S. development followed this iterative structure, the specific complexity of mapping 28 data stores and resolving NP-hard scheduling constraints necessitated an expanded prototype cycle of approximately 100 days to ensure total system reliability.

The architectural complexity of educational timetabling—which requires resolving NP-hard mathematical constraints while simultaneously providing an intuitive user interface—demands this highly iterative approach. Recent literature focusing on the automation of academic timetables strongly supports the use of the RAD model to navigate these specific challenges. Research indicates that RAD is uniquely suited for constraint-based scheduling software because it prioritizes structural flexibility, rapid adaptability to evolving institutional requirements, and the successful delivery of complex modules (Techie-Menson & Nyagorme, 2021).

By adopting this proven methodology, the development of A.T.L.A.S. prioritized iterative prototyping over rigid planning. This allowed the researchers to rapidly construct the core generation algorithm and the drag-and-drop visual interface, present these functional prototypes to the administrators at the participating pilot school, and immediately integrate their feedback. Consequently, the RAD approach ensured that the final system continuously evolved to meet the practical, real-world constraints of the end-users without causing severe delays in the development schedule.

Figure 1.

Rapid Application Development



a. Analysis and Quick Design

The researchers directly collaborated with the beneficiary to introduce the proposed system and gather essential insights to formulate the objectives and scope of the study. In this phase, the focus was on gathering qualitative information to establish an initial development plan, which served as the foundational basis for the system's architecture. Furthermore, the school's current manual workflow for creating timetables was analyzed to identify the significant parameters required for designing the automated timetabling algorithm, specifically subject allocations, room capacities, and major distributions.

b. Prototype Cycle (Build, Demonstrate, Refine)

After gathering the qualitative data, the development team started building the initial visual skeleton and functional prototype of the scheduling module. This early model was immediately shown to the beneficiary, specifically the school administrators and Master Teachers, to gather their direct feedback on how the system operated. The researchers then used this feedback to carefully improve the backend logic and user interface. This established the core repeating loop of the phase: building, demonstrating, and refining. This continuous cycle was repeated until the automated timetabling algorithm fully met the research objectives and perfectly matched the end-users' actual workflow.

c. Testing

Once the prototype cycles are finished, the developed system is ready for testing. First, the system will undergo Alpha Testing, where the researchers will rigorously test the software to find and fix any hidden bugs or errors. Afterward, the system will undergo Beta Testing. During this stage, the actual beneficiary, specifically the Master Teachers and administrators, will test the system using real school data to find any remaining issues. To evaluate how well the software solves their scheduling problems, the users will answer the Test Case Questionnaire. Finally, to measure the overall quality of the software, the researchers will use the standardized ISO 25010 questionnaire.

d. Implementation

The final step of the methodology is the Implementation phase, where A.T.L.A.S. (Automated Timetabling and Locations Allocation System) will be officially deployed at the designated beneficiary. To ensure high system availability and eliminate dependency on external internet service providers, the system will be deployed using a localized on-premises architecture and will not be hosted via cloud-based web hosting or a VPS.

The beneficiary institution will allocate a dedicated laptop or small on-prem server to act as the central local host for the system. This local host will use a Linux operating system and run the complete A.T.L.A.S. stack, including the web application and its supporting backend services, with a local PostgreSQL database for data persistence. End-users will access the web application through the school Local Area Network (LAN), allowing the system to operate at high speed within the campus network without requiring connectivity to the wider internet for core scheduling operations.

Following the technical installation, the researchers will conduct comprehensive user training sessions for administrative personnel, schedulers, and teacher. Training will emphasize how to configure the priority management module and how to use the drag-and-drop interface for fine adjustments during manual schedule refinement. Teacher will also be shown how to securely submit schedule preferences and view their synchronized schedules on mobile devices while connected to the school network. By establishing this localized, self-sustaining monolithic infrastructure, the goal is to make scheduling more efficient and manageable without recurring cloud hosting costs or latency associated with internet-based operations.

Research Respondents

The research respondents who are involved in this study are divided into two distinct groups. First group is tasked with system functionality testing, while the second group is task for the software quality evaluation.

The system functionality testing focuses in evaluating the core features outlined in the study's objectives, namely the web-based administrative portal and drag-and-drop interface (Objective 1.1), the configurable priority management module (Objective 1.2), the mobile teacher authentication and preference submission (Objective 1.3), and the automated timetable generation system (Objective 1.4). For this phase, the researcher purposively selected five (5) IT professionals who would act as major participants. These participants were selected for their specialized expertise in the software architecture, application development, and system analysis, ensuring that the quality of the evaluation will be aligned and be validated in both academic and industry standards.

Table 2

Test Case Respondent's Profile

Respondents	Age	Sex	Occupation	Device Used	Experience Level
R1	45	Female	IS Teacher	Android Phone	Advanced
R2	40	Female	IS Teacher	Laptop	Intermediate
R3	42	Male	IS Teacher	Laptop	Intermediate
R4	38	Male	IT Teacher	Android Phone	Advanced
R5	37	Male	IT Teacher	Laptop	Advanced

Based on the data presented in Table 2, the group for functionality testing are composed of IT teacher members and industry technical experts. These respondents possess high level of proficiency in technology-based learning. As shown in the table, 60%

of the respondents possess an advanced level of experience, while the remaining 40% are classified at an intermediate level. Their collective expertise ensures a highly reliable technical evaluation of the system's backend and logic components.

For the second group that will be responsible for the software quality evaluation using the ISO/IEC 25010 model, it will consist of actual end-users from the beneficiary school. There will be eight (8) Head Teachers, eight (8) Master Teachers, two (2) Principal, and seven (7) Regular Teachers.

These specific stakeholders were selected because of their understanding of academic scheduling and administrative processes and workflows. The Principal oversees the policy alignment of the schedules; the Head teacher inspects the set schedules of the Master teachers; the Master teacher organizes the time-tabling and scheduling of the teacher, student, subject, facilities and class; and the Regular Teacher utilizes the resulting schedules via mobile-responsive dashboard. Their real-world input will be highly valuable in verifying that the system maintains the institutional logic required in the scheduling flow by the Department of Education.

Table 3

Respondent's Profile for ISO/IEC 25010 Evaluation

Respondent Category	No. of Respondents	Percent
School Principals	2	8%
Master Teachers & Head Teachers (Schedulers)	16	64%
Regular Teachers	7	28%
TOTAL	25	100%

Table 3 illustrates the distribution of respondents for the software quality evaluation using the ISO/IEC 25010 model. There is a total of 25 respondents that participated in the evaluation. The largest group of respondents comprises the primary schedulers, which is the Head and Master Teachers, at 64%, followed by Regular Teachers at 28%, and the School Principal at 8%. This distribution will ensure the quality of the system evaluated not only from the administrative perspective in overseeing the scheduling policy, but also by the end-users who will interact with the application daily. The involvement of the specific stakeholders will allow the researchers to acquire comprehensive feedbacks on the system's usability and functional usability for real-world secondary school environment.

Sampling Technique

The study utilizes non- probability purposive sampling for the selection of the respondents. According to Rai and Thapa (2015), purposive sampling relies on the judgement of the researcher when selecting the units to be studied. It is a targeted approach where decisions concerning the individuals to be included in the sample are based upon a variety of criteria, specifically including their specialist knowledge of the research issue. Because this study focuses on complex academic scheduling conflicts and evaluating software architecture, it was strictly necessary to rely on this methodology to select IT professionals and administrative schedulers. Their specialized technical expertise ensures that the system is evaluated by individuals with the exact capacity required to understand the application's underlying logic.

Once these specific experts were selected, the study employed a strict quantitative evaluation approach. As defined by Lim (2024), quantitative research concentrates on the collection of measurable variables and quantifiable data to objectively analyze phenomena and arrive at reliable conclusions. By having the purposively selected experts evaluate the A. T. L. A. S. system using the numerical 5-point Likert scale of the ISO/IEC 25010 model and strict binary pass-or-fail test cases, the researchers successfully gathered objective, statistically valid data to measure the software's overall performance without relying on subjective opinions

For the Functionality Testing, the respondents were purposively selected base on their backgrounds on how technical they are in software development and IT infrastructure. This guarantees a rigorous quality testing of the core features of the system, specifically the automated timetable generation and the configurable priority management module.

For the Software Quality Evaluation, the respondents will be the end-user purposively selected from the beneficiary school. This includes the principal, head scheduler, and regular teachers. Their deep, practical understanding of the school's administrative sequence and DepEd scheduling rules will be significantly necessary to ensure that the feedback on the system's functionality and usability is base on the real-life administrative experiences of local Junior High Schools. This approach guarantees a highly accurate and institutionally valid assessment of the system's strengths and weaknesses.

Data Gathering Instrument

The researcher will utilize a research-made Test Case Questionnaire for assessing the algorithm's operational logic and the ISO/IEC 25010:2011 Questionnaire for evaluating the overall software product quality. These data gathering instruments will help comprehensively evaluate the Automated Timetabling and Locations Allocation System (A.T.L.A.S.)

Test Case Questionnaire. To ensure that each core technical feature of A. T. L. A. S. functions correctly and meets the specified system requirement, the researchers developed a standardized functional test case document. Instead of utilizing a subjective observational checklist, this testing instrument strictly follows the established AAA (Arrange, Act,Assert) software engineering principle. This ensures the system is rigorously tested against both expected user behaviors (positive testing) and unexpected or invalid inputs (negative testing).

The Test Case Questionnaire consists of ten (10) distinct testing scenarios that are aligned with the study's objectives. These scenarios evaluate the system's critical functionalities, specifically Admin Authentication and Drag-and Drop Interface, the Priority Management Module, Teacher Authentication and Preference Submission, and the Automated Timetable Generation Engine. Furthermore, the performance of each test case is objectively measure using binary status. A scenario is marked as "Pass" only if the actual system response perfectly matches the predefined expected result, ensuring strict functional reliability prior to the system's deployment. Table 4 presents the verbal interpretations, providing a clearer understanding of how each system function performed during evaluation.

Table 4

Test Case Result with Verbal Interpretation

Pass Rate (%)	Verbal Interpretation	Description
96-100%	Excellent	The system fulfills its functional requirement with very few problems.
90-95%	Very Good	The system operates effectively but has some areas that could be enhanced.
85-89%	Good	The system is operational but needs several adjustments.
80-84%	Fair	The system has significant issues that impact its performance.
Below 80%	Poor	The system does not fulfill most of its intended functions.

ISO/IEC 25010:2011 Questionnaire. A local research of Keh and Sarmiento (2025) utilized the ISO 25010 industry standard questionnaire for evaluating their automated scheduling system, and it demonstrated an efficiency in testing the software product quality of the system. It measures the quality attributes such as functional suitability, performance efficiency, and usability, which will allow researchers to quantitatively verify that a scheduling system successfully streamlines operations and will meet the administrative expectations. Consequently, A.T. L. A. S. will adopt the ISO 25010 framework for its evaluation phase. This standardized metric will be used to systematically validate the system's reliability, security, and operational effectiveness during its deployment.

The standardized questionnaire will consist of thirty-one (31) questions measuring eight (8) primary software characteristics, which are Functional Suitability (3 items), Performance Efficiency (3 items), Usability (6 items), Compatibility (2 items), Reliability (4 items), Security (5 items), Maintainability (5 items), and Portability (3 items). Table 5 presents the verbal interpretations, providing a clearer understanding of how each system function performed during evaluation.

Table 5

5-point Likert Scale with Verbal Interpretation

Numerical Code	Interval	Verbal Interpretation	Description
5	4.51 – 5.00	Excellent	The system is complete and fully functional.
4	3.51 – 4.50	Very Good	The system is complete and functional.
3	2.51 – 3.50	Good	The system is somewhat complete and functional.
2	1.51 – 2.50	Poor	The system is incomplete and somewhat functional.
1	1.50 – below	Very Poor	The system is incomplete and non-functional.

Data Gathering Procedure

The data gathering process started upon securing an official written approval from the Dean of the College of Computer Studies at Carlos Hilado Memorial State University – Alijis Campus. Then, the researchers, together with the Capstone Adviser and the Instructor, conducted an online formal meeting with the identified academic institutional beneficiary. Following this, the researcher drafted formal letters of acceptance confirming the beneficiary's agreement to be included in the study.

Once approved, the researchers conducted requirements interview and presented an initial skeleton prototype to the beneficiary via face-to-face demonstration. Through this, the researchers determined the existing gaps in the manual process and the preferences of the beneficiary, instructor and the adviser. Because the study utilized the Rapid Application

Development (RAD) methodology, data gathering then transitioned into a continuous, parallel task. While the researcher rebuilt and improve the system logic during development iterations, they maintained continuous communication with the beneficiary to gather ongoing operational information and integrated technical feedback from the Capstone Adviser during scheduled technical prototype reviews.

After the A.T.L.A.S. (Automated Timetabling and Locations Allocation System was fully built, formal data collection for software evaluation will follow. The researchers created a formal letter for the five (5) IT professionals in agreement to participate in test case validation in their available time. Through purposive sampling, these experts will execute a structured functional test cases (Alpha Testing) to determine the technical robustness, reliability, and performance level of the system in various environment. During this phase, the researchers will directly observe and record the system's success rate in handling specific data-driven parameters, specifically spatial constraints, human constraints, and operational logic.

Following the functionality tests, the researchers will provide survey respondents with enough time to explore their assigned portals before they begin their evaluation of the application's functional suitability and performance efficiency and usability. Once familiarized, the standardized ISO/IEC 25010:2011 Questionnaire having thirty-one (31) items will be distributed to the School Administrators, Master Teachers, and regular teachers to evaluate the complete software product quality. Following this, the feedback that had been acquired from IT professionals and end-users will be compiled, organized and then stored for analysis. With this outcome, a comprehensive statistical analysis will be conducted to determine the overall quality of the system and identify final areas of improvement.

Validity of Research Instrument

The data gathering instruments utilized for evaluating the software product quality is the standardized ISO/IEC 25010:2011 questionnaire. Because it is recognized as an international standard for evaluating software quality, its validity is already established and accepted globally.

Reliability of the Questionnaire

The ISO/IEC 25010:2011 standard is a globally accepted and highly reliable tool for measuring software systems. Local studies, such as the research conducted by Keh and Sarmiento (2025), have successfully utilized this exact framework to reliably evaluate automated class scheduling systems within the Philippine educational context.

Analysis/Statistical Treatment of Data

A weighted mean and a success rate percentage will be applied as the primary statistical tools to systematically quantify and analyze the data gathered during the evaluation phases of A.T.L.A.S. These tools allow the researchers to calculate scores that reflect the central tendency of the evaluators' responses across different grading scales.

For the functional test case evaluation, the success rate will be used to quantify the algorithmic performance based on binary (Pass/Fail) results. This provides an objective measure of system reliability. The formula for computing the success rate is:

$$\text{Test Case Success Rate} = \frac{\text{Number of Passed Test Cases}}{\text{Total Number of Test Cases}} \times 100$$

This metric is used to analyze the operational logic of the algorithm. Scores from IT professionals are averaged per individual testing scenario to determine success rates on specific rules, such as room double-bookings. These scenario means are then grouped to provide a specialized look at which logistical areas handle constraints perfectly and which require further mathematical refinement.

Similarly, a systematic aggregation process will be followed for the ISO/IEC 25010 system evaluation to determine the overall software product quality. The mean will be used to quantify the evaluators' responses gathered from the 5-point Likert scale. The individual performance of every sub-characteristic will be measured first to identify specific strengths and weaknesses in the system. These initial scores will then be grouped and averaged according to their specific ISO quality characteristic, such as Functional Suitability or Performance Efficiency, providing a clear assessment of different technical domains. Ultimately, these characteristic means will be averaged to calculate the final numerical value representing the total quality and deployment readiness of the web application.

The mathematical formula utilized for calculating the mean is represented as follows

$$\bar{x} = \frac{\sum x}{N}$$

Where $\bar{x}$ represents the calculated mean or average score, $\sum x$ represents the total sum of all the scores or ratings provided by the respondents, and N represents the total number of evaluators who participated in the testing phase.

Ethical Consideration

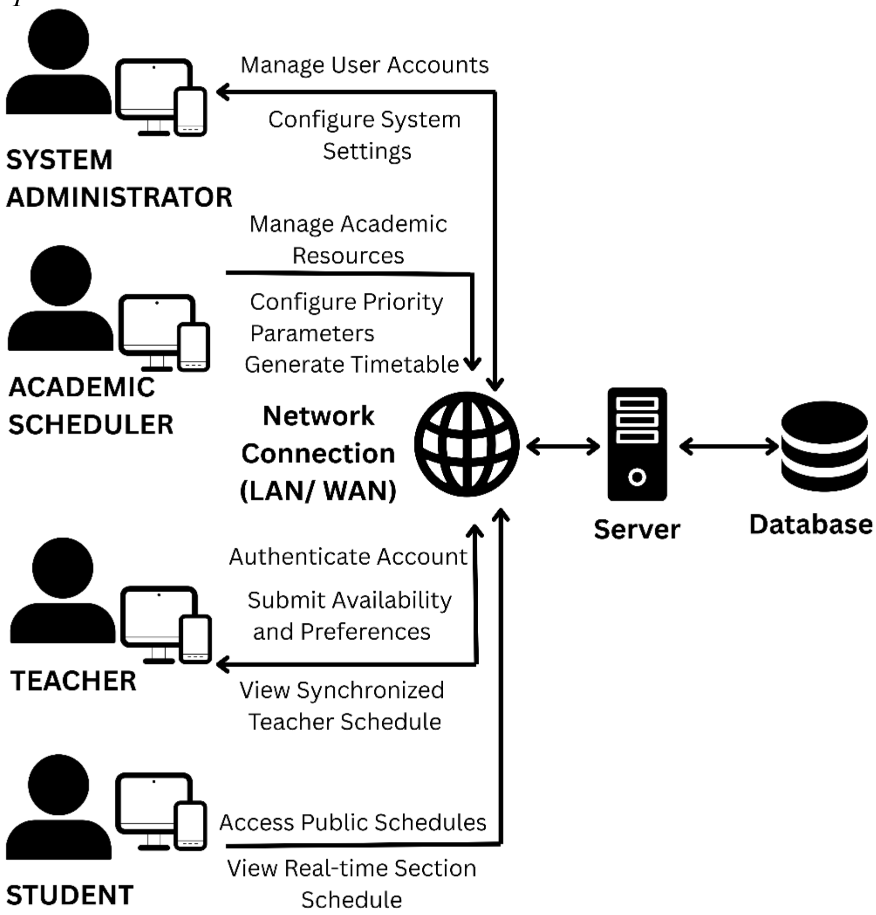
This study strictly complied with the established ethical standards and considerations to ensure that the rights and well-being of the respondents who participated were protected and respected. Informed consent was distributed and obtained from each respondent to ensure that they understood the purpose, the procedure, and the risks involved in their participation. The participation in this study was completely voluntary, and each respondent had the right to withdraw without any consequences. It is important to note that any personal or institutional data provided by the respondents during the evaluation of the system was handled with strict adherence to data privacy guidelines. The confidentiality of their data was strictly maintained and was not shared with any other parties without their consent. All the feedback that was gathered was solely used to improve A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application, and remained inside the institution of Carlos Hilado Memorial State University.

Operational Framework

Figure 2 illustrates the operational framework of A.T.L.A.S., depicting the live interaction between the four primary actors and the system infrastructure. The flow begins with the System Administrator and Academic Scheduler, who provide administrative configurations and core resource data. These inputs are transmitted via a local or wide area network connection to the centralized Server, where the scheduling logic is processed and persisted within the Database. The framework then facilitates the distribution of synchronized outputs, allowing Teachers to manage preferences and view their specific schedules, while Students access real-time section updates through the public web interface.

Figure 2.

Operational Framework

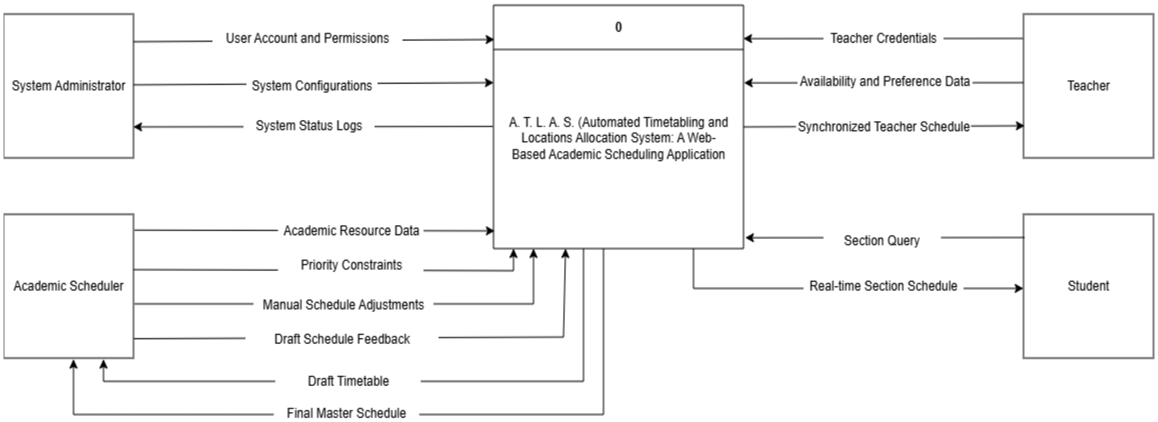


Context Diagram (Level 0)

Figure 3 depicts the A.T.L.A.S. Context Diagram, outlining the high-level data exchange between the system and its four key entities. The System Administrator manages user accounts and permissions while monitoring system status logs to ensure technical integrity. The Academic Scheduler provides resource data, priority constraints, and manual adjustments to receive iterative drafts and finalized master schedules. Teachers submit credentials, availability, and preferences to access their synchronized teacher schedules. Finally, Students retrieve real-time updates via open-access section queries. This architecture prioritizes high accessibility by utilizing the school's local and wide-area network (LAN/WAN) to allow the student population to view schedules without authentication, ensuring immediate access to time-sensitive academic information even during public internet interruptions.

Figure 3.

Context Diagram (Level 0)



Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram (DFD) provides a functional decomposition of the A.T.L.A.S. system, detailing the logical movement of information between eight core processes and 28 primary data stores. To enhance readability and reduce visual complexity, the system is organized into three functional clusters that utilize layman-friendly noun phrases to describe specific data packets. This architecture ensures that every administrative action, from institutional data synchronization to final schedule publication, remains consistent with the system's centralized database and technical documentation.

Figure 4 illustrates the foundational security and synchronization processes of the system. Process 1.0 (User Authentication) manages access by verifying saved account credentials against the authentication store and logging every login attempt to the audit trail for security monitoring.

Process 2.0 (Manage System Settings) handles the initial configuration of school profiles and subject specialization rules. It is also responsible for the critical synchronization of teacher and section data, comparing external snapshots with local mirror records to ensure the system utilizes the most current institutional data before any scheduling begins.

Figure 4

Data Flow Diagram (Level 1.1 – Identity & System Basics)

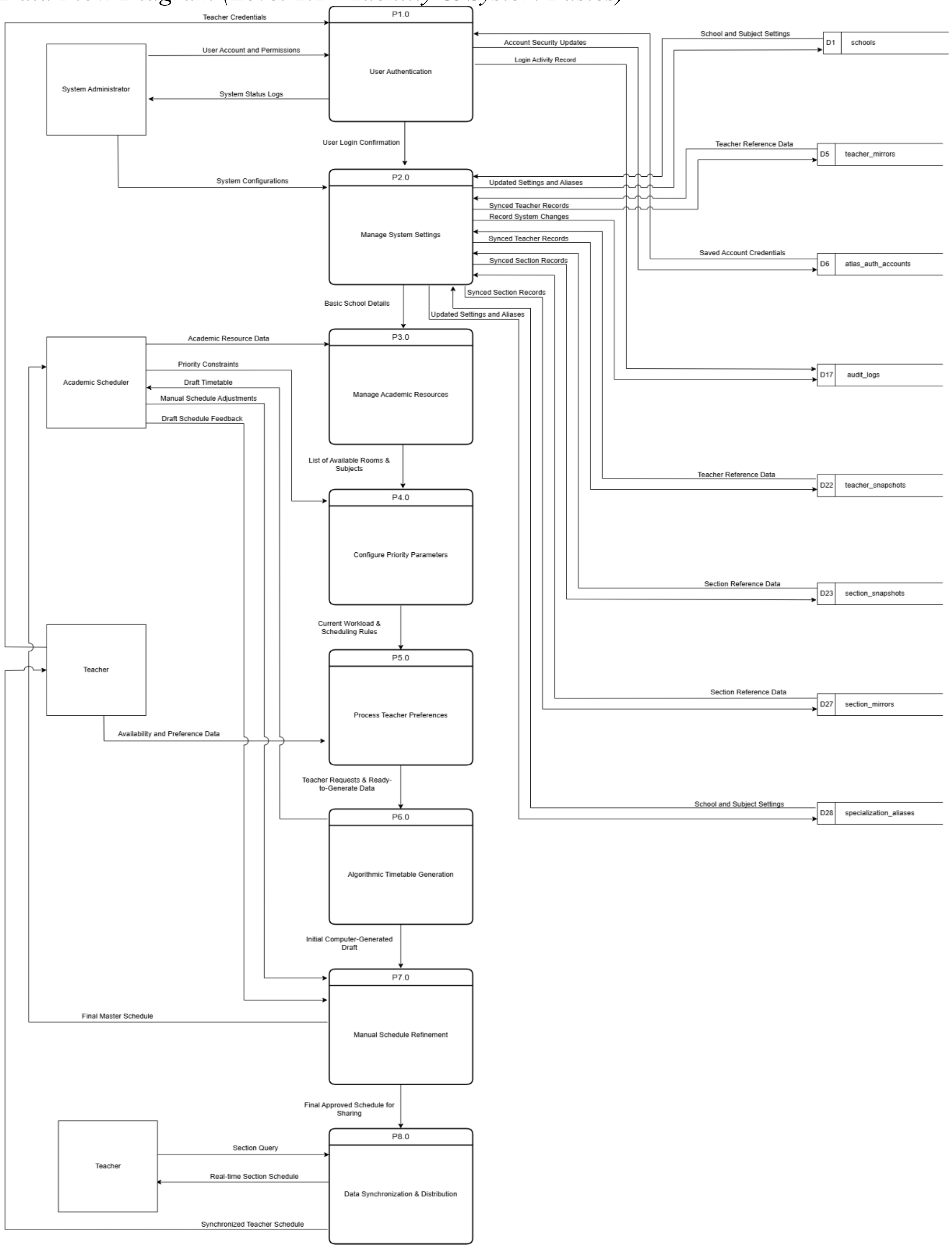


Figure 5 details how the system organizes the physical and human constraints required for scheduling. Process 3.0 (Manage Academic Resources) allows the scheduler to maintain a comprehensive school resource list, including buildings, rooms, subjects, and specific teacher-subject assignments.

Process 4.0 (Configure Priority Parameters) establishes the core scheduling rules and grade-level shift windows that guide the automation engine. Finally, Process 5.0 (Process Teacher Preferences) captures direct input from the teaching staff, managing submitted preference forms, review outcomes, and any subsequent room request appeals or escalations.

Figure 5

Data Flow Diagram (Level 1.2 – Resource & Preference)

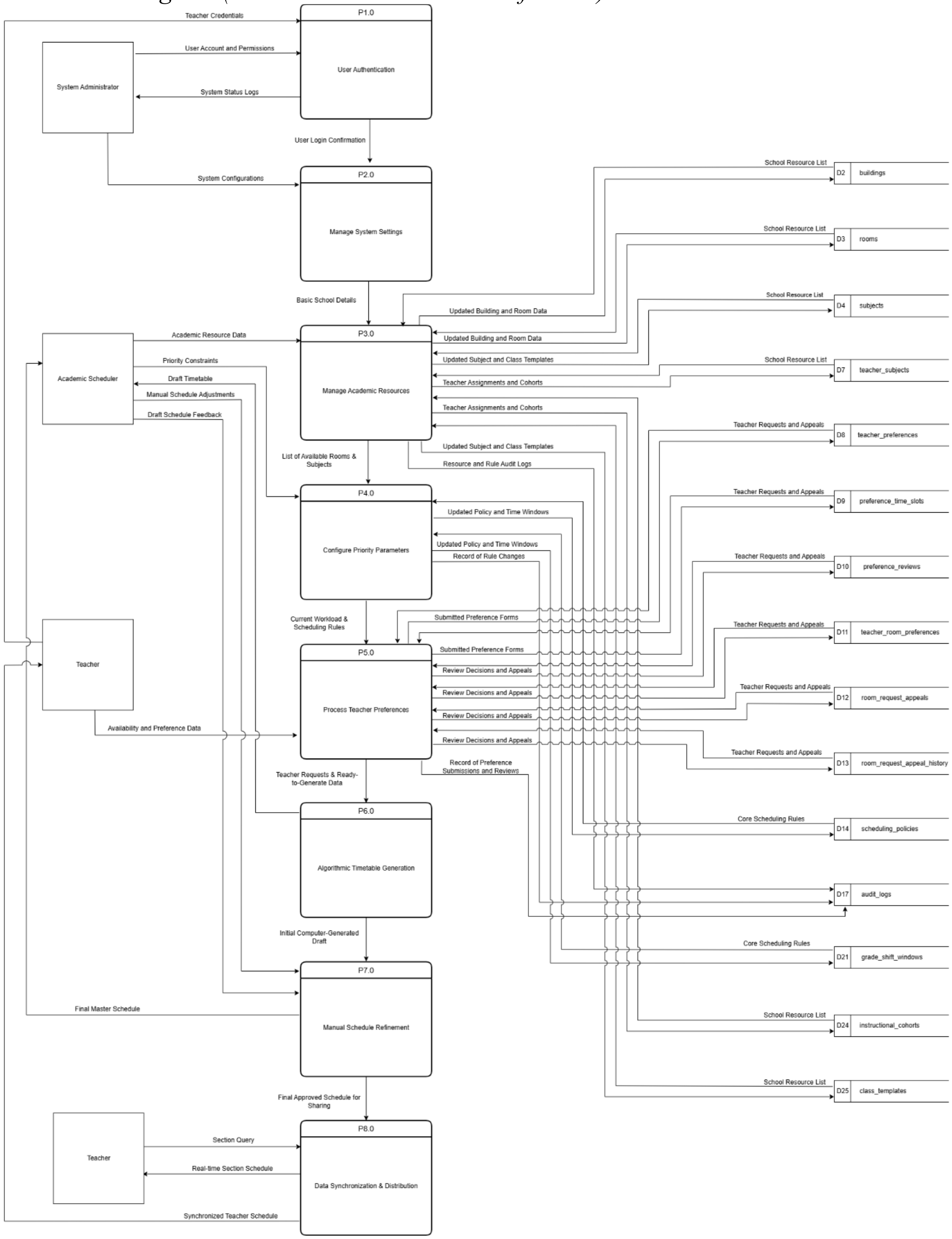
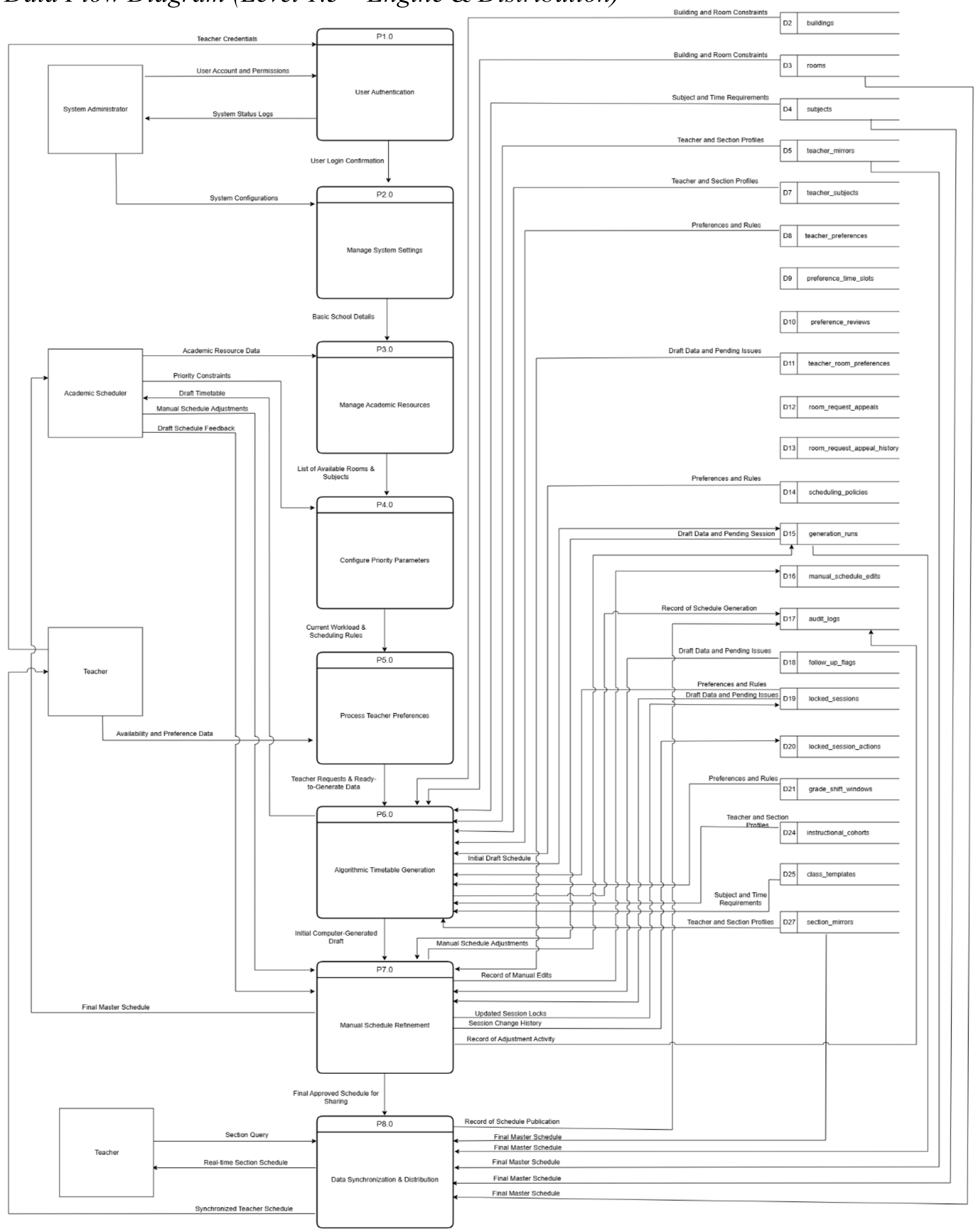


Figure 6 presents the "production" phase of the system. Process 6.0 (Algorithmic Timetable Generation) acts as the core engine, pulling together teacher load profiles, subject requirements, room constraints, and established rules to create an initial computer-generated draft.

Process 7.0 (Manual Schedule Refinement) provides the scheduler with the tools to perform manual adjustments, allowing for the pinning of specific sessions and the tracking of all revision history. Once fine-tuned, Process 8.0 (Data Synchronization and Distribution) combines the final master schedule with enriched teacher and room context to publish a synchronized timetable for the entire school community.

Figure 6

Data Flow Diagram (Level 1.3 – Engine & Distribution)

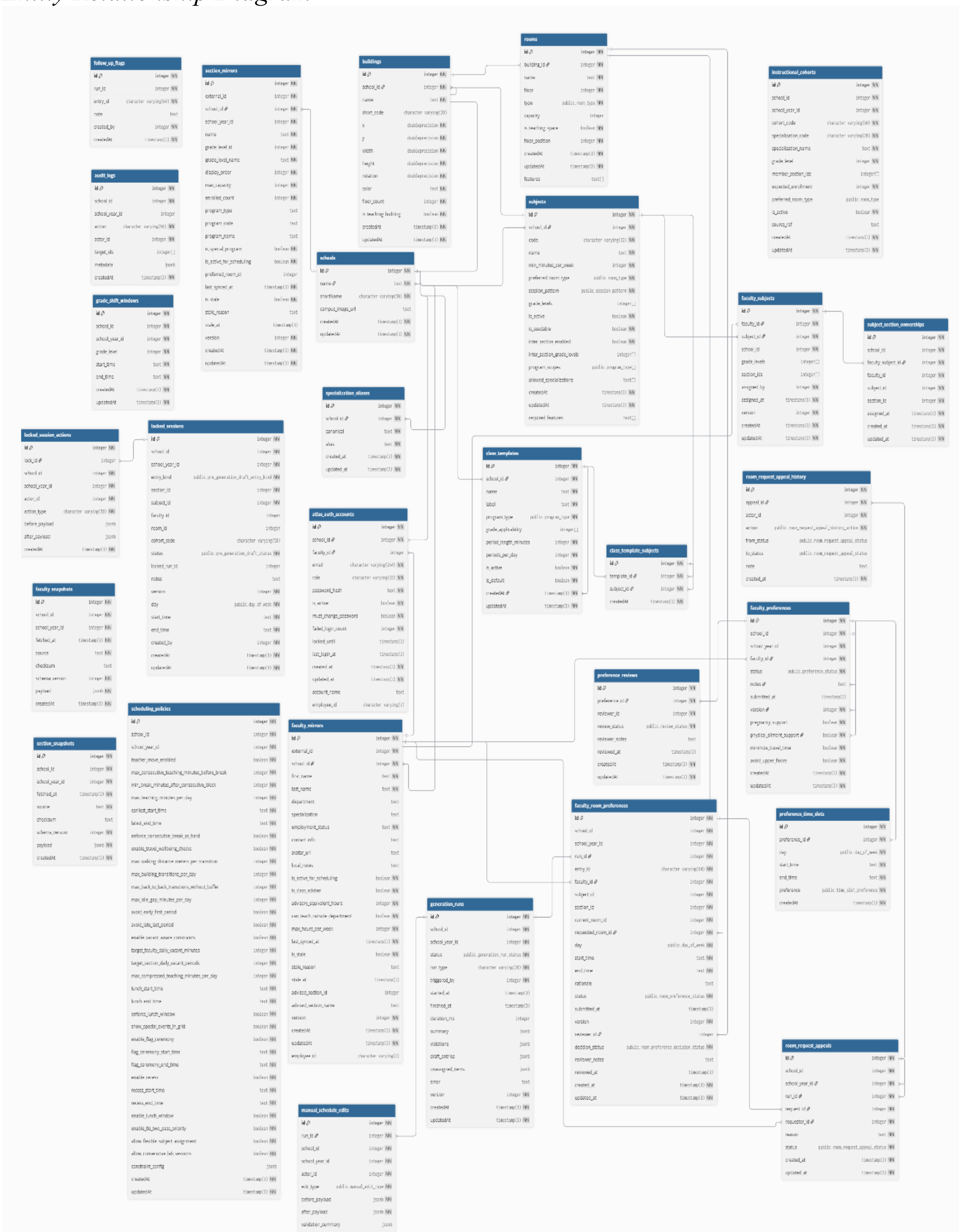


Entity Relationship Diagram

Figure 7 illustrates the relational connections between entities within the Automated Timetabling and Locations Allocation System (A.T.L.A.S.) database. This entity relationship diagram (ERD) ensures organized data management by mapping the system's logical structure, specifically linking institutional foundations like schools and buildings with academic resources and teacher preferences. By providing a comprehensive view of the 28-table architecture, the ERD allows developers and researchers to easily grasp the complex dependencies that drive the automated scheduling algorithm.

Figure 7.

Entity Relationship Diagram

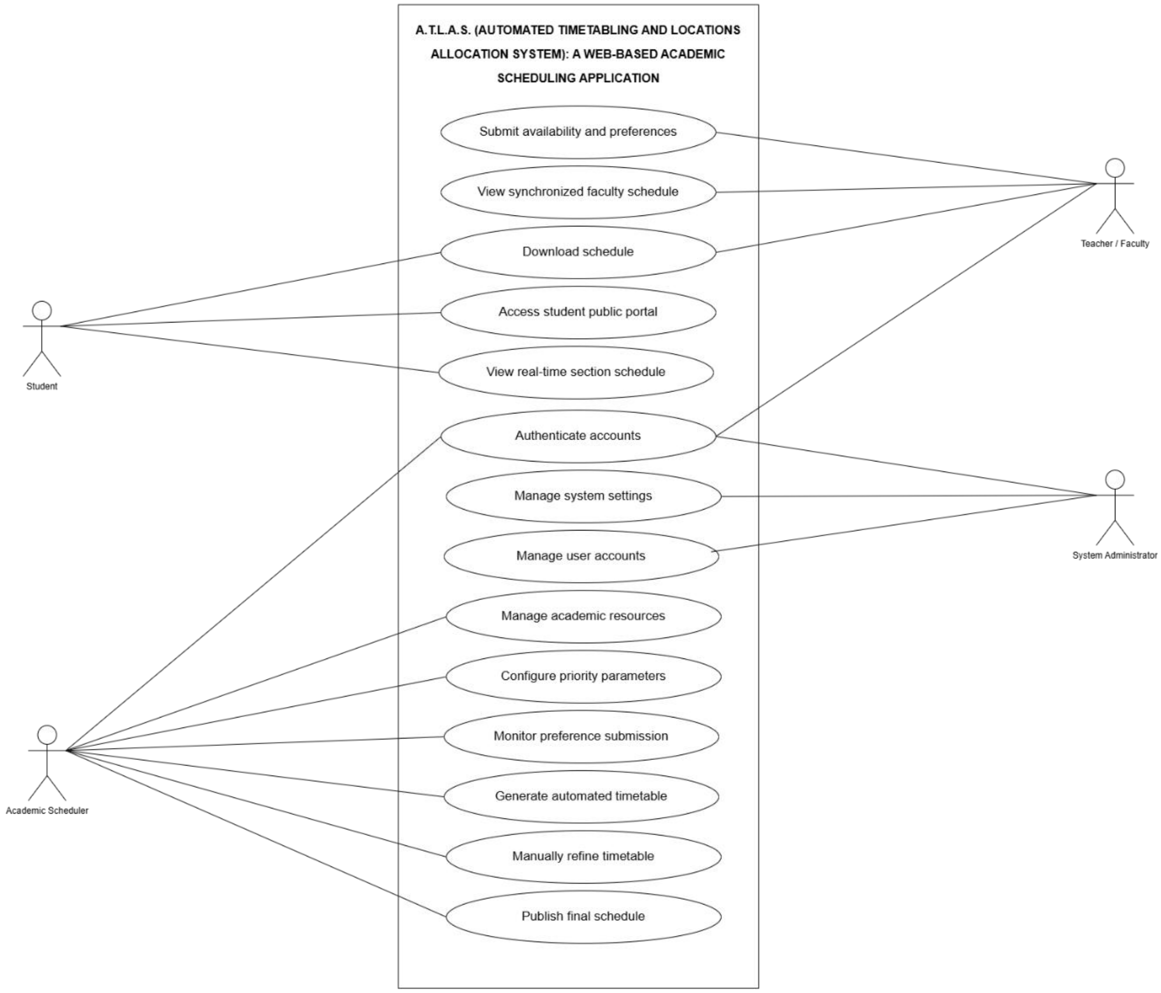


Use Case Diagram

Figure 8 illustrates the user interactions within A.T.L.A.S. by identifying four primary actors and their specific role-based privileges. The System Administrator maintains technical integrity by managing user accounts and securing credentialed access for school personnel. The Academic Scheduler serves as the primary operator by managing academic resources, configuring priority parameters, and triggering the automated generation engine. Teachers utilize a secure portal to submit availability data and view their synchronized teacher schedules. Finally, students access the open-access schedule page to view real-time section updates and download class timetables without the need for an authenticated account.

Figure 8.

Use Case Diagram



Data Dictionary

The data dictionary is a component that stores the names, definitions, and attributes for data elements present inside the database of the proposed system. The tables listed below, starting from Table 6, provide the listings of the actual field names, data types, lengths, and descriptions of data elements that are present in the database of A.T.L.A.S.

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Table 6

Schools Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the school.
name	VARCHAR	255	Full legal name of the school.
shortName	VARCHAR	50	Short name or abbreviation for the school.
campus_image_url	VARCHAR	2048	Optional; URL of the campus map image.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 6 shows foundational profile information for each school institution, including the school id, the full legal name, the short name or abbreviation, the campus image URL, and the timestamps for when the record was created and updated.

Table 7

Buildings Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the building.
school_id	INT (FK)	---	Reference to the owning school
name	VARCHAR	100	Building name.
short_code	VARCHAR	20	Optional; short building code.
x	FLOAT	---	X coordinate on the campus map.
y	FLOAT	---	Y coordinate on the campus map.
width	FLOAT	---	Width of the building on the map.
height	FLOAT	---	Height of the building on the map.
rotation	FLOAT	---	Rotation angle for map rendering.
color	VARCHAR	7	Hex color used for map rendering.
floor_count	INT	---	Number of floors.
is_teaching_building	BOOLEAN	---	True when the building contains teaching spaces.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 7 shows the details of campus buildings, including the building id, the school reference id, the building name, the short code, the map coordinates and dimensions, the rotation angle, the rendering color, the floor count, and whether it is a teaching building, alongside the record creation and update timestamps.

Table 8

Rooms Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the room.
building_id	INT (FK)	---	Reference to the building containing the room.
name	VARCHAR	50	Room name or number.
floor	INT	---	Floor number where the room is located.
type	ENUM	---	Room type. One of: CLASSROOM, LABORATORY, COMPUTER_LAB, TLE_WORKSHOP, LIBRARY, GYMNASIUM, FACULTY_ROOM, OFFICE, OTHER.
capacity	INT	---	Optional; maximum occupancy capacity.
is_teaching_space	BOOLEAN	---	True when the room can be scheduled for classes.
floor_position	INT	---	Ordering position of the room on its floor.
features	VARCHAR	---	List of room feature tags (e.g., PROJECTOR, AC). Defaults to empty array
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 8 shows the details of specific rooms within the buildings, including the room id, the building reference id, the room name or number, the floor level, the room type, the occupancy capacity, whether it is a teaching space, the floor position, the list of room features, and the record creation and update timestamps.

Table 9

Subjects Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the subject.
school_id	INT (FK)	---	Reference to the owning school.
code	VARCHAR	32	Subject code (unique within the school).
name	VARCHAR	100	Subject name.
min_minutes_per_week	INT	---	Required minimum instructional minutes per week.

Field Name	Field Type	Length	Description
preferred_room_type	ENUM	---	Preferred room type. One of: CLASSROOM, LABORATORY, COMPUTER_LAB, TLE_WORKSHOP, LIBRARY, GYMNASIUM, FACULTY_ROOM, OFFICE, OTHER.
session_pattern	ENUM	---	Preferred session distribution pattern. One of: MWF (Mon/Wed/Fri), TTH (Tue/Thu), ANY (no preference).
grade_levels	INT	---	Grade levels this subject applies to.
is_active	BOOLEAN	---	True when the subject is active and usable.
is_seedable	BOOLEAN	---	True when subject can be included in seed data sets.
inter_section_enabled	BOOLEAN	---	True when inter-section grouping is allowed.
inter_section_grade_levels	INT	---	Grade levels allowed for inter-section grouping.
program_scopes	ENUM	---	Program types this subject applies to. Values: REGULAR, STE, SPS, SPA, OTHER.
allowed_specializations	VARCHAR	---	Teacher specializations allowed to teach the subject.
required_features	VARCHAR	---	Room feature tags required by the subject (e.g., PROJECTOR). Defaults to empty array.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 9 shows the academic subjects offered, including the subject id, the school reference id, the subject code, the subject name, the required instructional minutes, the preferred room type and session pattern, the applicable grade levels, the required room features, and various status flags for active and groupable settings, as well as the record creation and update timestamps.

Table 10

Teacher Mirrors Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique local identifier for a teacher member.
external_id	INT	---	External LIS/HR identifier.
employee_id	VARCHAR	7	Optional; unique DepEd employee ID (e.g., "1234567").
school_id	INT (FK)	---	Reference to the owning school.
first_name	VARCHAR	100	Teacher first name.
last_name	VARCHAR	100	Teacher last name.
department	VARCHAR	100	Optional; department name.
specialization	VARCHAR	100	Optional; specialization name.
employment_status	VARCHAR	50	Employment status label (e.g., PERMANENT).

Field Name	Field Type	Length	Description
contact_info	VARCHAR	255	Optional; contact details.
avatar_url	VARCHAR	2048	Optional; profile image URL.
local_notes	VARCHAR	1000	Optional; school-specific notes.
is_active_for_scheduling	BOOLEAN	---	True when included in scheduling.
is_class_adviser	BOOLEAN	---	True when assigned as a class adviser.
advisory_equivalent_hours	INT	---	Advisory hours credited in workload.
can_teach_outside_department	BOOLEAN	---	True when allowed to teach outside department.
max_hours_per_week	INT	---	Maximum teaching hours per week.
last_synced_at	DATETIME	---	Last sync time with LIS/HR source.
is_stale	BOOLEAN	---	True when mirror data is stale.
stale_reason	VARCHAR	255	Optional; reason for staleness.
stale_at	DATETIME	---	Optional; time the record was marked stale.
advised_section_id	INT	---	Optional; section ID advised.
advised_section_name	VARCHAR	100	Optional; cached advised section name.
version	INT	---	Revision counter for this record.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 10 shows the local records of teacher members, including the local and external identifiers, the specific employee id, the school reference id, personal and professional details like name and department, employment and advisory status, workload constraints, and synchronization metadata, including the creation and update timestamps.

Table 11

ATLAS Authentication Accounts Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the auth account.
school_id	INT (FK)	---	Reference to the owning school.
teacher_id	INT (FK)	---	Optional; linked teacher mirror ID.
email	VARCHAR	254	Unique login email.
employee_id	VARCHAR	7	Optional; unique DepEd employee ID linked to the account.
account_name	VARCHAR	---	Optional; unique display/username for the account.
role	VARCHAR	32	Role label for access control.
password_hash	VARCHAR	---	Hashed password.
is_active	BOOLEAN	---	True when the account is active.
must_change_password	BOOLEAN	---	True when a password reset is required on next login.
failed_login_count	INT	---	Consecutive failed login attempts.
locked_until	DATETIME	---	Optional; lockout expiration time.
last_login_at	DATETIME	---	Optional; last successful login time.
created_at	DATETIME	---	Record creation timestamp.
updated_at	DATETIME	---	Record last update timestamp.

Table 11 shows the authentication account details, including the account id, the school and teacher reference ids, the login email and employee id, the unique account name, the access control role, the hashed password, the active status, the password reset flag, login security metrics, and the record creation and update timestamps.

Table 12

Teacher Subjects Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the assignment.
teacher_id	INT (FK)	---	Linked teacher member.
subject_id	INT (FK)	---	Linked subject.
school_id	INT (FK)	---	Reference to the owning school.
grade_levels	INT	---	Grade levels covered by the assignment.
section_ids	INT	---	Section IDs assigned under this pairing.
assigned_by	INT	---	ID of the staff who made the assignment.
assigned_at	DATETIME	---	Assignment timestamp.
version	INT	---	Revision counter for this record.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 12 shows the assignments between teachers and their respective subjects, including the assignment id, the teacher and subject identifiers, the school reference id, the assigned grade levels and section ids, and information on who made the assignment and when it was recorded.

Table 13

Teacher Preferences Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the preference form.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
teacher_id	INT (FK)	---	Submitting teacher member.

Field Name	Field Type	Length	Description
status	ENUM	---	Preference form status. One of: DRAFT, SUBMITTED.
notes	VARCHAR	1000	Optional; teacher notes or constraints.
submitted_at	DATETIME	---	Optional; submission timestamp.
version	INT	---	Revision counter for this record.
pregnancy_support	BOOLEAN	---	Indicates pregnancy-related support request.
physical_ailment_support	BOOLEAN	---	Indicates physical ailment accommodation request.
minimize_travel_time	BOOLEAN	---	Indicates preference to reduce travel time.
avoid_upper_floors	BOOLEAN	---	Indicates preference to avoid upper floors.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 13 shows the preference forms submitted by teachers, including the preference id, the school and teacher reference ids, the school year context, the submission status, specialized support or accommodation requests, and the timestamps for when the record was created and updated.

Table 14

Preference Time Slots Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the time slot.
preference_id	INT (FK)	---	Reference to the teacher preference record.
day	ENUM	---	Day of week. One of: MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY.
start_time	VARCHAR	5	Start time in HH:MM format.
end_time	VARCHAR	5	End time in HH:MM format.
preference	ENUM	---	Preference rating for the time slot. One of: PREFERRED, AVAILABLE, UNAVAILABLE.
createdAt	DATETIME	---	Record creation timestamp.

Table 14 shows the specific time intervals for teacher availability, including the time slot id, the reference to the teacher preference record, the day of the week, the start and end times in HH:MM format, the preference rating for that specific slot, and the record creation timestamp.

Table 15

Preference Reviews Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the review.
preference_id	INT (FK)	---	Reference to the teacher preference record.
reviewer_id	INT	---	Reviewer identifier.
review_status	ENUM	---	Review status. One of: PENDING, REVIEWED, NEEDS_FOLLOW_UP.
reviewer_notes	VARCHAR	1000	Optional; reviewer notes.
reviewed_at	DATETIME	---	Optional; time review was completed.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 15 shows the review details for teacher preference submissions, including the review id, the reference to the teacher preference record, the identifier of the reviewer, the current review status, any notes provided by the reviewer, the timestamp of when the review was completed, and the record creation and update timestamps.

Table 16

Teacher Room Preferences Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the room preference.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
run_id	INT (FK)	---	Reference to the generation run.
entry_id	VARCHAR	64	Entry identifier from the generation draft.
teacher _id	INT (FK)	---	Teacher member requesting the change.
subject_id	INT (FK)	---	Subject tied to the request.
section_id	INT	---	Section identifier tied to the request.
current_room_id	INT	---	Current assigned room ID.
requested_room_id	INT	---	Requested room ID.
day	ENUM	---	Day of week. One of: MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY.
start_time	VARCHAR	5	Start time in HH:MM format.
end_time	VARCHAR	5	End time in HH:MM format.
rationale	VARCHAR	---	Optional; justification for the request.
status	ENUM	---	Submission status. One of: DRAFT, SUBMITTED.
submitted_at	DATETIME	---	Optional; time request was submitted.
version	INT	---	Revision counter for this record.
reviewer_id	INT	---	Optional; reviewer identifier.
decision_status	ENUM	---	Decision status. One of: PENDING, APPROVED, REJECTED.

Field Name	Field Type	Length	Description
reviewer_notes	VARCHAR	1000	Optional; reviewer notes.
reviewed_at	DATETIME	---	Optional; review completion time.
created_at	DATETIME	---	Record creation timestamp.
updated_at	DATETIME	---	Record last update timestamp.

Table 16 shows the requests made by teachers for specific room assignments, including the room preference id, the school and school year reference ids, the generation run and entry identifiers, the requesting teacher and subject ids, the section id, the current and requested room ids, the requested day and time, the rationale for the request, the submission and decision status, reviewer details and notes, and the record creation and update timestamps.

Table 17

Room Request Appeals Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the appeal.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
run_id	INT (FK)	---	Reference to the generation run.
request_id	INT (FK)	---	Reference to the room preference request.
requester_id	INT (FK)	---	Teacher member who filed the appeal.
reason	VARCHAR	1000	Stated reason for the appeal.
status	ENUM	---	Appeal status. One of: OPEN, UNDER_REVIEW, UPHELD, DENIED.
created_at	DATETIME	---	Record creation timestamp.
updated_at	DATETIME	---	Record last update timestamp.

Table 17 shows the appeals filed regarding room assignment decisions, including the appeal id, the school and school year reference ids, the generation run and room request reference ids, the identifier of the teacher filing the appeal, the stated reason for the appeal, the current appeal status, and the record creation and update timestamps.

Table 18

Room Request Appeal History Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for a history entry.
appeal_id	INT (FK)	---	Reference to the appeal.
actor_id	INT	---	Identifier of the actor who made the change.
action	ENUM	---	Action taken. One of: CREATED, STATUS_CHANGED, NOTE_ADDED, DECISION_RECORDED.
from_status	ENUM	---	Optional; previous appeal status. One of: OPEN, UNDER_REVIEW, UPHELD, DENIED.
to_status	ENUM	---	Optional; new appeal status. One of: OPEN, UNDER_REVIEW, UPHELD, DENIED.
note	VARCHAR	1000	Optional; free-form note.
created_at	DATETIME	---	Record creation timestamp.

Table 18 shows the historical logs of room request appeals, including the history entry id, the reference to the appeal, the identifier of the actor who made the change, the specific action taken, the previous and new statuses, optional notes, and the record creation timestamp.

Table 19

Scheduling Policies Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the policy.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
teacher_move_enabled	BOOLEAN	---	Allows teachers to move between rooms.
max_consecutive_teaching_minutes_before_break	INT	---	Maximum consecutive teaching minutes before break.
min_break_minutes_after_consecutive_block	INT	---	Minimum break minutes after a block.
max_teaching_minutes_per_day	INT	---	Maximum teaching minutes per day.
earliest_start_time	VARCHAR	5	Earliest start time in HH:MM format.
latest_end_time	VARCHAR	5	Latest end time in HH:MM format.

Field Name	Field Type	Length	Description
enforce_consecutive_break_as_hard	BOOLEAN	---	Treat missing break as a hard constraint.
enable_travel_wellbeing_checks	BOOLEAN	---	Enable travel and wellbeing constraints.
max_walking_distance_meters_per _transition	INT	---	Max walking distance per transition.
max_building_transitions_per_day	INT	---	Max building transitions per day.
max_back_to_back_transitions _without_buffer	INT	---	Max back-to-back transitions without buffer.
max_idle_gap_minutes_per_day	INT	---	Max idle gap minutes per day.
avoid_early_first_period	BOOLEAN	---	Avoid early first period for teacher.
avoid_late_last_period	BOOLEAN	---	Avoid late last period for teacher.
enable_vacant_aware_constraints	BOOLEAN	---	Enable vacant-aware constraints.
target_ teacher _daily_vacant_minutes	INT	---	Target daily vacant minutes for teacher.
target_section_daily_vacant_periods	INT	---	Target daily vacant periods for sections.
max_compressed_teaching_minutes _per_day	INT	---	Max compressed teaching minutes per day.
lunch_start_time	VARCHAR	5	Lunch window start time in HH:MM format.
lunch_end_time	VARCHAR	5	Lunch window end time in HH:MM format.
enforce_lunch_window	BOOLEAN	---	Enforce lunch window as a constraint.
show_special_events_in_grid	BOOLEAN	---	Display special events in schedule grids.
enable_flag_ceremony	BOOLEAN	---	Enable flag ceremony time block.
flag_ceremony_start_time	VARCHAR	5	Flag ceremony start time in HH:MM format.
flag_ceremony_end_time	VARCHAR	5	Flag ceremony end time in HH:MM format.
enable_recess	BOOLEAN	---	Enable recess time block.
recess_start_time	VARCHAR	5	Recess start time in HH:MM format.
recess_end_time	VARCHAR	5	Recess end time in HH:MM format.
enable_lunch_window	BOOLEAN	---	Enable lunch window time block.
enable_tle_two_pass_priority	BOOLEAN	---	Enable two-pass priority for TLE scheduling.
allow_flexible_subject_assignment	BOOLEAN	---	Allow flexible subject assignment rules.
allow_consecutive_lab_sessions	BOOLEAN	---	Allow consecutive lab sessions.

Field Name	Field Type	Length	Description
constraint_config	JSON	---	Optional; JSON blob for additional constraint config.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 19 shows the global scheduling policies and configuration settings, including the policy id, school and school year references, teacher movement and workload limits, temporal constraints like start and end times, travel and wellbeing check parameters, transition limits, vacancy targets for teachers and sections, meal and ceremony window configurations, specialized subject and lab rules, and the general constraint configuration, alongside the record creation and update timestamps.

Table 20

Generation Runs Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the generation run.
school_id	INT (PK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
status	ENUM	---	Run status. One of: QUEUED, RUNNING, COMPLETED, FAILED.
run_type	VARCHAR	20	Run type label (e.g., FULL).
triggered_by	INT	---	Actor identifier who triggered the run.
started_at	DATETIME	---	Optional; run start timestamp.
finished_at	DATETIME	---	Optional; run end timestamp.
duration_ms	INT	---	Optional; run duration in milliseconds.
summary	JSON	---	Optional; run summary payload.
violations	JSON	---	Optional; constraint violations payload.
draft_entries	JSON	---	Optional; draft schedule entries payload.
unassigned_items	JSON	---	Optional; unassigned items payload.
error	VARCHAR	500	Optional; failure error message.
version	INT	---	Revision counter for this record.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 20 shows the execution logs for automated scheduling runs, including the run id, school and school year references, current run status and type, the actor who triggered the process, temporal metadata for start and finish times, the total duration, detailed summaries and violation reports, result payloads for drafts and unassigned items, failure messages, and the record versioning and creation timestamps.

Table 21

Manual Schedule Edits Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the manual edit.
run_id	INT (FK)	---	Reference to the generation run.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
actor_id	INT	---	Actor identifier who made the edit.
edit_type	ENUM	---	Type of manual edit. One of: PLACE_UNASSIGNED, MOVE_ENTRY, CHANGE_ROOM, CHANGE_FACULTY, CHANGE_TIMESLOT, SWAP_ENTRIES, REVERT.
before_payload	JSON	---	Snapshot before the edit.
after_payload	JSON	---	Snapshot after the edit.
validation_summary	JSON	---	Optional; validation summary after the edit.
createdAt	DATETIME	---	Record creation timestamp.

Table 21 shows the records of manual adjustments applied to generated schedules, including the edit id, the generation run and institutional references, the identifier of the actor who performed the edit, the specific type of adjustment, snapshots of the schedule data before and after the modification, the post-edit validation summary, and the record creation timestamp.

Table 22

Audit Logs Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the audit entry.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	Optional; school year context identifier.
action	VARCHAR	50	Action label.
actor_id	INT	---	Actor identifier for the action.
target_ids	INT	---	Target entity IDs associated with the action.

Field Name	Field Type	Length	Description
metadata	JSON	---	Optional; metadata payload.
createdAt	DATETIME	---	Record creation timestamp.

Table 22 shows the system's audit trail, including the audit entry id, the school and optional school year reference ids, the specific action performed, the identifier of the actor responsible, the list of target entity IDs, additional metadata, and the record creation timestamp.

Table 23

Follow Up Flags Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the follow-up flag.
run_id	INT (FK)	---	Reference to the generation run.
entry_id	VARCHAR	64	Entry identifier in the draft schedule.
note	VARCHAR	500	Optional; follow-up note.
created_by	INT	---	Actor identifier who created the flag.
createdAt	DATETIME	---	Record creation timestamp.

Table 23 shows the flags created for manual follow-up in the schedule, including the flag id, the generation run reference id, the specific entry identifier in the draft, optional notes, the identifier of the creator, and the record creation timestamp.

Table 24

Locked Sessions Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the locked session.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
entry_kind	ENUM	---	Entry type. One of: SECTION, COHORT.
section_id	INT	---	Section identifier.
subject_id	INT	---	Subject identifier.
teacher_id	INT	---	Optional; teacher identifier.
room_id	INT	---	Optional; room identifier.
cohort_code	VARCHAR	50	Optional; cohort code for grouped sessions.
status	ENUM	---	Draft status. One of: DRAFT, LOCKED_FOR_RUN, ARCHIVED.
locked_run_id	INT	---	Optional; generation run that locked the entry.
notes	VARCHAR	500	Optional; notes for the locked session.

Field Name	Field Type	Length	Description
version	INT	---	Revision counter for this record.
day	ENUM	---	Day of week. One of: MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY.
start_time	VARCHAR	5	Start time in HH:MM format.
end_time	VARCHAR	5	End time in HH:MM format.
created_by	INT	---	Actor identifier who created the lock.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 24 shows the details of sessions locked within the draft, including the locked session id, school and school year references, the entry kind, section and subject identifiers, the teacher and room ids, the cohort code, the draft status, the locking run id, optional notes, the record version, the day and time interval, the identifier of the creator, and the record creation and update timestamps.

Table 25

Locked Session Actions Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the lock action.
lock_id	INT (FK)	---	Optional; reference to the locked session.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
actor_id	INT	---	Actor identifier who performed the action.
action_type	VARCHAR	50	Action type label.
before_payload	JSON	---	Optional; payload before action.
after_payload	JSON	---	Optional; payload after action.
createdAt	DATETIME	---	Record creation timestamp.

Table 25 shows the history of actions performed on locked sessions, including the action id, the reference to the locked session, school and school year references, the actor identifier, the type of action performed, snapshots of the payload before and after the action, and the record creation timestamp.

Table 26

Grade Shift Windows Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the grade shift window.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
grade_level	INT	---	Grade level the window applies to.
start_time	VARCHAR	5	Window start time in HH:MM format.
end_time	VARCHAR	5	Window end time in HH:MM format.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 26 shows the time windows defined for specific grade levels, including the shift window id, the school and school year reference ids, the applicable grade level, the window start and end times, and the record creation and update timestamps.

Table 27

Teacher Snapshots Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the snapshot.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
fetched_at	DATETIME	---	Snapshot fetch timestamp.
source	VARCHAR	100	Source system label.
checksum	VARCHAR	64	Optional; checksum of the payload.
schema_version	INT	---	Snapshot schema version.
payload	JSON	---	Snapshot payload.
createdAt	DATETIME	---	Record creation timestamp.

Table 27 shows the historical snapshots of teacher data retrieved from external sources, including the snapshot id, the school and school year reference ids, the fetch timestamp, the source system label, the optional checksum, the schema version, the payload content, and the record creation and update timestamps.

Table 28

Section Snapshots Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the snapshot.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
fetched_at	DATETIME	---	Snapshot fetch timestamp.
source	VARCHAR	100	Source system label.
checksum	VARCHAR	64	Optional; checksum of the payload.
schema_version	INT	---	Snapshot schema version.
payload	JSON	---	Snapshot payload.
createdAt	DATETIME	---	Record creation timestamp.

Table 28 shows the snapshots of student section data retrieved from external systems, including the snapshot id, the school and school year reference ids, the fetch timestamp, the source system label, the optional checksum, the schema version, the payload content, and the record creation and update timestamps.

Table 29

Instructional Cohorts Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the cohort.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
cohort_code	VARCHAR	50	Cohort code identifier.
specialization_code	VARCHAR	20	Specialization code.
specialization_name	VARCHAR	100	Specialization name.
grade_level	INT	---	Grade level.
member_section_ids	INT	---	Section IDs in the cohort.
expected_enrollment	INT	---	Expected total enrollment.
preferred_room_type	ENUM	---	Optional; preferred room type. One of: CLASSROOM, LABORATORY, COMPUTER_LAB, TLE_WORKSHOP, LIBRARY, GYMNASIUM, FACULTY_ROOM, OFFICE, OTHER.
is_active	BOOLEAN	---	True when the cohort is active.
source_ref	VARCHAR	100	Optional; source reference label.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 29 shows the details of instructional cohorts used for grouping student sections, including the cohort id, the school and school year reference ids, the cohort and specialization codes, the specialization name, the grade level, the member section ids, the expected enrollment, the preferred room type, the active status, and the record creation and update timestamps.

Table 30

Class Templates Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the class template.
school_id	INT (FK)	---	Reference to the owning school.
name	VARCHAR	100	Template name.
label	VARCHAR	100	Display label for the template.
program_type	ENUM	---	Program type (ProgramType enum).
grade_applicability	INT	---	Grade levels this template applies to.
period_length_minutes	INT	---	Length of each period in minutes.
periods_per_day	INT	---	Number of periods per day.
is_active	BOOLEAN	---	True when the template is active.
is_default	BOOLEAN	---	True when the template is the default.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 30 shows the templates for class structures, including the template id, the school reference id, the template name and display label, the program type, the applicable grade levels, the period length in minutes, the number of periods per day, the active and default status flags, and the record creation and update timestamps.

Table 31

Class Template Subjects Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the template binding.
template_id	INT (FK)	---	Reference to the class template.
subject_id	INT (FK)	---	Reference to the subject.
createdAt	DATETIME	---	Record creation timestamp.

Table 31 shows the relationship between class templates and subjects, including the template binding id, the reference to the class template, the reference to the subject, and the record creation timestamp.

Table 32

Section Mirrors Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the section mirror record.
external_id	INT	---	External LIS/EnrollPro section identifier.
school_id	INT (FK)	---	Reference to the owning school.
school_year_id	INT	---	School year context identifier.
name	VARCHAR	---	Section name (e.g., "7-Rizal").
grade_level_id	INT	---	Numeric grade level identifier from the source system.
grade_level_name	VARCHAR	---	Display name for the grade level (e.g., "Grade 7").
display_order	INT	---	Ordering position within the grade level.
max_capacity	INT	---	Maximum enrollment capacity of the section.
enrolled_count	INT	---	Current enrolled student count.
program_type	VARCHAR	---	Optional; program type label from the source system.
program_code	VARCHAR	---	Optional; program code from the source system.
program_name	VARCHAR	---	Optional; program name from the source system.
is_special_program	BOOLEAN	---	True when the section belongs to a special program.
is_active_for_scheduling	BOOLEAN	---	True when the section is included in ATLAS scheduling.
preferred_room_id	INT	---	Optional; ATLAS-assigned preferred room override.
last_synced_at	DATETIME	---	Last sync time with the LIS/EnrollPro source.
is_stale	BOOLEAN	---	True when the mirror data is stale relative to the source.
stale_reason	VARCHAR	---	Optional; reason the record was marked stale.
stale_at	DATETIME	---	Optional; time the record was marked stale.
version	INT	---	Revision counter for this record.
createdAt	DATETIME	---	Record creation timestamp.
updatedAt	DATETIME	---	Record last update timestamp.

Table 32 shows the local mirrors of section data from external systems, including the mirror id, the external section identifier, the school and school year reference ids, the section name and grade level details, the display order, the enrollment capacity and counts, the program type information, the active and stale status flags, and the record creation and update timestamps.

Table 33

Specialization Aliases Table

Field Name	Field Type	Length	Description
id	INT (PK)	---	Unique identifier for the alias entry
school_id	INT (FK)	---	Reference to the owning school.
canonical	VARCHAR	---	Canonical specialization name (e.g., "Mathematics").
alias	VARCHAR	---	Accepted alias for the canonical name (e.g., "Algebra").
created_at	DATETIME	---	Record creation timestamp.
updated_at	DATETIME	---	Record last update timestamp.

Table 33 shows the mappings for specialization aliases, including the alias entry id, the school reference id, the canonical specialization name, the accepted alias, and the record creation and update timestamps.

Hardware and Software Requirement

Hardware Requirements

The hardware requirements encompass the high-performance workstations used during the development phase and the local server dedicated to hosting the application. By utilizing a self-hosted server architecture, the institution maintains full control over its academic data while eliminating the recurring costs of cloud-based web hosting.

Table 34.

A.T.L.A.S. Hardware Requirements Summary

Components	Minimum Specification	Recommended Specification	Justification
Server CPU	AMD Ryzen 5 (3000 Series+) or Intel Core i5 (10th Gen+)	Intel Xeon or AMD EPYC (12+ Cores)	Supports concurrent timetable generation, conflict validation, and API request handling for multi-role users (scheduler,

			faculty, and student viewers).
Server RAM	8 GB DDR4	16 GB to 32 GB DDR4/DDR5	Maintains active scheduling sessions, policy evaluation workloads, and PostgreSQL processing during peak usage periods.
Server Storage	256 GB HDD / SSD	1 TB NVMe SSD (RAID 1 Configuration)	Stores timetable datasets, generation artifacts, audit logs, integration records, and routine system backups.
Network (ISP)	25 Mbps stable Symmetric connection	100 Mbps to 1 Gbps Symmetric Fiber	Enables dependable on-campus access via LAN for core scheduling tasks even without public internet, while providing secure off-campus access when internet connectivity is available
Scheduler/Admin Client	Intel Core i3 (8th Gen) or AMD Ryzen 3, 8 GB RAM	Intel Core i5 (11th Gen) or AMD Ryzen 5, 12 GB RAM	Supports timetable review tools, drag-and-drop adjustments, and administrative dashboard workflows.
Teacher Client (Mobile/ PC)	Intel Core i3 (8th Gen) or AMD Ryzen 3, 8 GB RAM	Intel Core i5 (11th Gen) or AMD Ryzen 5, 12 GB RAM	Ensures smooth use of faculty preference submission and schedule viewing features on mobile-responsive interfaces.
Student Client (Mobile/ PC)	Android 10.0+ / iOS 15.0+, 4 GB RAM	Android 13.0+ / iOS 17.0+, 6 GB RAM	Ensures compatibility with the student view-only schedule portal and modern browser rendering requirements.

Software Requirements

To successfully deploy A.T.L.A.S. within the institution's integrated Enterprise Resource Planning (ERP) platform, the dedicated on-premises server must run a stable and secure software stack for continuous school operations. The recommended server operating system is Ubuntu Server LTS 22.04 or higher. PostgreSQL (version 16 or higher) is required as the primary database engine, while Prisma ORM manages structured data access and schema migrations. Node.js v20 LTS is required to run the backend services, scheduling logic, and related background processes.

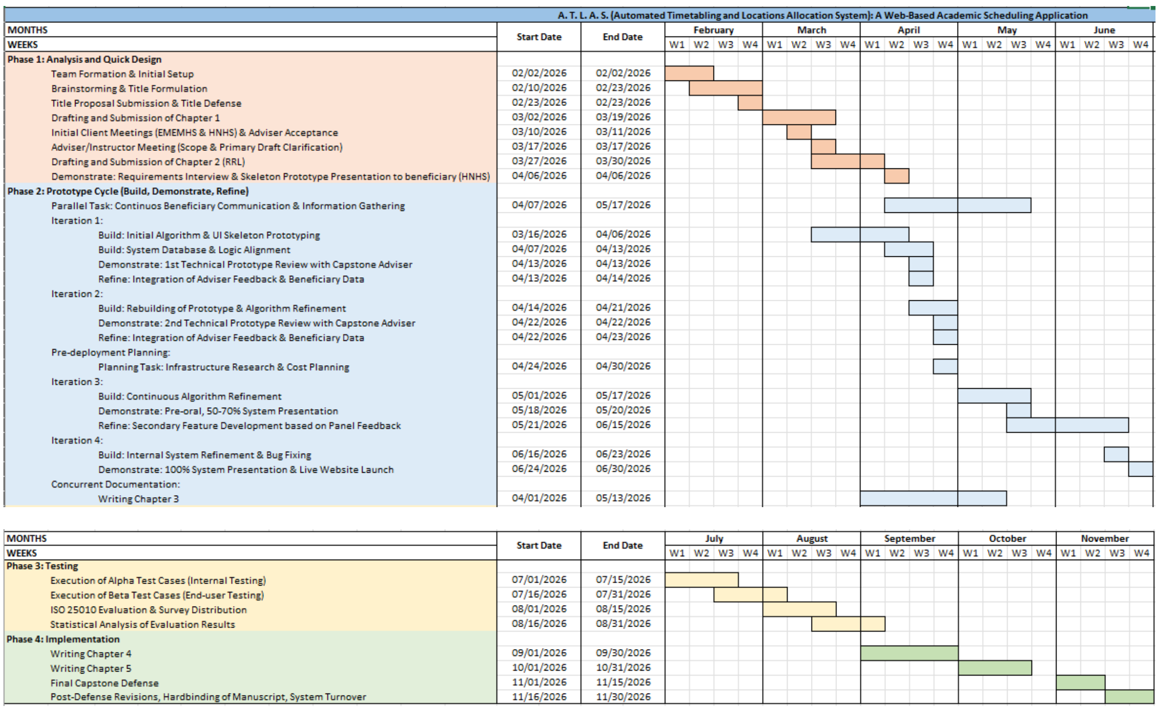
For reliability and maintainability, Docker and Docker Compose v2.x are recommended to containerize services with clear process isolation and predictable resource use, especially in constrained hardware environments. Nginx (version 1.24+) is recommended as a reverse proxy and static asset gateway to improve request handling and reduce application server load. Security controls include JWT-based authentication, bcrypt password hashing, and role-based access enforcement for academic scheduler, teacher users, and system administrator. For release management, Git and an automated testing/deployment workflow should be used so updates and fixes can be delivered consistently with minimal downtime. Remote support tools may be installed subject to institutional IT policy. On the client side, no specialized installation is required beyond a modern browser (Chrome, Edge, Firefox, or Safari) capable of accessing the ERP-hosted web portals through the school LAN or approved secure remote channel.

Gantt Chart

Figure 9 presents the project schedule through a Gantt chart, detailing a ten-month timeline from February to November 2026. The development process is divided into four distinct phases to ensure a systematic workflow. Phase 1 (Analysis and Quick Design) covers initial requirements gathering and manuscript development. Phase 2 (Prototype Cycle) involves the iterative building and refinement of the A.T.L.A.S. algorithm based on technical reviews. Phase 3 (Testing) encompasses rigorous alpha and beta evaluations using functional test cases and ISO/IEC 25010 standards. Finally, Phase 4 (Implementation) concludes with the final defense, manuscript revisions, and the official system turnover in late November.

Figure 9.

A. T. L. A. S. Gantt Chart



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Appendices

APPENDIX A

Transmittal Letters

Adviser Acceptance Form

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IT CAPSTONE PROJECT ADVISER ACCEPTANCE FORM

Date: March 10, 2026

This is to attest that I, **Jayrelle B. Sy** have agreed and accepted to be the **Adviser** of the following students with the Systems Project entitled ***"A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application."***

Group Name: **CGG**
Course/Yr./Sec: **BSIT 3 - B**

Members:

1. **CURIO, JOSH NATHAN I.**
2. **GILERA, ROWENA G.**
3. **GROMEA, NEHJE JOHN S.**

Furthermore, I agree to set the schedule for advising and consultation to help the students and ensure the success of the project.

Conformed:

JAYRELLE B. SY, Ph.D.
Adviser's Signature over Printed Name

Noted by:

MARITES D. MANGANTI, LPT, Ph.D.
IT Capstone Project Professor

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CAPSTONE PROJECT ADVISER ROLES AND RESPONSIBILITIES
BOT Resolution No. 44 Series of 2013 /Revised 2020 Capstone Project Manual

The adviser must have completed an IT/IS project successfully beyond the bachelor's degree project. As much as possible, the adviser should be a full-time faculty member of the HEI (Higher Education Institution) otherwise a full-time faculty co-adviser is required.

The following are the duties and responsibilities of the Capstone Project adviser:
1. **Overall supervision of research of the students.** The participation of the adviser starts at the proposal stage up to the manuscript final revision stage. The adviser should regularly confer and meet his or her advisees. The adviser may suggest the following, but not limited to, research topic, framework, and methodology.
2. **Checking errors and required corrections of the manuscript.** To be fair to the adviser, the checking of errors is limited to technical details and not grammatical errors. However, the adviser can give the students the directions on the proper format of the study.
3. **Technical supervision in the writing of the research proposal**. The adviser should be physically present during the proposal and final defense but is not allowed to participate in the defense unless allowed by the panel members.
4. **Setting the deadlines for their advisees.** The deadline is within the discretion of the adviser but must be bounded with the deadline of Capstone Project Instructor; however, paramount consideration should be made regarding institutional deadlines. For formality and consistency, all manuscripts should be ready for final defense 15 days before the end of the first semester or earlier should the student wish to defend his/her
thesis in advance. The adviser should evaluate first whether the manuscript is ready for defense. If so, the adviser should request the Seniors System Project Professor/Instructor for a schedule of final defense.
5. **Maintain contact and coordination with the advisees until the submission of the revised final manuscript.** In the final revision stage, the adviser should duly accomplish the routing slip every time the manuscript comes in and out of his/her responsibility.
6. **Become a member of the group as a co-proponent.** In case the Capstone Project is selected as an outstanding study and or be published in a journal, the adviser will be credited as a "co-proponent" for his/her contributions upon the approval of the group, the research instructor and of the Dean of the College. It would also be the adviser's responsibility to accompany the group in research presentations.

Conformed:

JAYRELLE B. SY, Ph.D.
Adviser's Signature over Printed Name

CERTIFICATION INTERNATIONAL ISO 9001:2015
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Consent Form

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PARENT/GUARDIAN WAIVER AND CONSENT FORM
Research/ Capstone Project

I, **Ma. Concepcion I. Curio**, the parent/guardian of **Josh Nathan I. Curio**, a student of Carlos Hilado Memorial State University, College of Computer Studies, Alijis Campus, currently enrolled in the Capstone Project course in BSIT Program, hereby give my full consent for my child/ward to conduct research/capstone project titled **"A. T. L. A. S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application" at Hinigaran National High School.**

I acknowledge that off-campus activities involve various research conditions, including, but not limited to, changes in the activity schedule, potential health risks, and transportation challenges. With full awareness of these conditions, I hereby declare that:

1. I hereby grant permission for my child/ward to participate in this activity and assume any associated risks.
2. I voluntarily allow my child/ward to conduct research activities, including system development, data gathering (interviews/observations), testing, and implementation, which may require working beyond regular hours. I understand that these activities will take place outside the school premises from 8:00 AM to 5:00 PM from **February 27, 2026, until June 30, 2027,** and may also extend beyond this schedule if necessary.
3. I acknowledge that transportation to and from the research site is my child/ward's responsibility.
4. I release Carlos Hilado Memorial State University – Alijis Campus, its faculty, staff, and administrators from any liability for incidents that may occur beyond their control during off-campus research/capstone project activities.
5. I commit to ensuring that my child/ward adheres to all beneficiary workplace policies, safety protocols, and any additional rules imposed by the beneficiary/company/organization regarding their research/ capstone project activities.
6. I have been informed of my child/ward's research/capstone project schedule and off-campus activities, and I hereby give my full consent for their participation.

By signing this waiver, I confirm that I am fully aware of the conditions and potential risks associated with these activities and that I accept full responsibility for my decision.

Parent/Guardian's Name : Ma. Concepcion I. Curio
Signature :
Date : March 12, 2026
Contact Number : 09063672084
Email Address : maconcepcion.curio@deped.gov.ph

Conforme:
Student's Name : Josh Nathan I. Curio
Signature :
Date : March 12, 2026

ENDORSEMENT FROM THE UNIVERSITY

Noted:

JAYRELLE W SY, Ph.D.
IT Capstone Project Adviser
E-mail: *jayrelle.sy@chmsu.edu.ph*

MARITES D. MANGANTI, LPT, Ph.D.
IT Capstone Project Professor/ BSIT Program Chairperson
E-mail: marites.manganti@chmsu.edu.ph

JOE MARIE D. DORMIDO, DIT, PCpE
Dean, College of Computer Studies
E-mail: joe.dormido@chmsu.edu.ph

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Research/ Capstone Project

I, **Florentina G. Gilera**, the parent/guardian of **Rowena G. Gilera**, a student of Carlos Hilado Memorial State University, College of Computer Studies, Alijis Campus, currently enrolled in the Capstone Project course in BSIT Program, hereby give my full consent for my child/ward to conduct research/capstone project titled **"A. T. L. A. S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application"** at **Hinigaran National High School.**

I acknowledge that off-campus activities involve various research conditions, including, but not limited to, changes in the activity schedule, potential health risks, and transportation challenges. With full awareness of these conditions, I hereby declare that:

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2. I voluntarily allow my child/ward to conduct research activities, including system development, data gathering (interviews/observations), testing, and implementation, which may require working beyond regular hours. I understand that these activities will take place outside the school premises from 8:00 AM to 5:00 PM from **February 27, 2026, until June 30, 2027,** and may also extend beyond this schedule if necessary.
3. I acknowledge that transportation to and from the research site is my child/ward's responsibility.
4. I release Carlos Hilado Memorial State University – Alijis Campus, its faculty, staff, and administrators from any liability for incidents that may occur beyond their control during off-campus research/capstone project activities.
5. I commit to ensuring that my child/ward adheres to all beneficiary workplace policies, safety protocols, and any additional rules imposed by the beneficiary/company/organization regarding their research/ capstone project activities.
6. I have been informed of my child/ward's research/capstone project schedule and off-campus activities, and I hereby give my full consent for their participation.

By signing this waiver, I confirm that I am fully aware of the conditions and potential risks associated with these activities and that I accept full responsibility for my decision.

Parent/Guardian's Name : Florentina G. Gilera
Signature : Gilera
Date : March 12, 2026
Contact Number : 09481382186
Email Address : florentinagilera@gmail.com

Conforme:
Student's Name : Rowena G. Gilera
Signature :
Date : March 12, 2026

ENDORSEMENT FROM THE UNIVERSITY

Noted:

JAYRELLE B. SY, Ph.D.
IT Capstone Project Adviser
E-mail: *jayrelle.sy@chmsu.edu.ph*

MARITES D. MANGANTI, LPT, Ph.D.
IT Capstone Project Professor/ BSIT Program Chairperson
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PARENT/GUARDIAN WAIVER AND CONSENT FORM
Research/ Capstone Project

I, **Julie Darlene S. Gromea**, the parent/guardian of **Nehje John S. Gromea**, a student of Carlos Hilado Memorial State University, College of Computer Studies, Alijis Campus, currently enrolled in the Capstone Project course in BSIT Program, hereby give my full consent for my child/ward to conduct research/capstone project titled **"A. T. L. A. S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application" at Hinigaran National High School.**

I acknowledge that off-campus activities involve various research conditions, including, but not limited to, changes in the activity schedule, potential health risks, and transportation challenges. With full awareness of these conditions, I hereby declare that:

1. I hereby grant permission for my child/ward to participate in this activity and assume any associated risks.
2. I voluntarily allow my child/ward to conduct research activities, including system development, data gathering (interviews/observations), testing, and implementation, which may require working beyond regular hours. I understand that these activities will take place outside the school premises from 8:00 AM to 5:00 PM from **February 27, 2026, until June 30, 2027,** and may also extend beyond this schedule if necessary.
3. I acknowledge that transportation to and from the research site is my child/ward's responsibility.
4. I release Carlos Hilado Memorial State University – Alijis Campus, its faculty, staff, and administrators from any liability for incidents that may occur beyond their control during off-campus research/capstone project activities.
5. I commit to ensuring that my child/ward adheres to all beneficiary workplace policies, safety protocols, and any additional rules imposed by the beneficiary/company/organization regarding their research/ capstone project activities.
6. I have been informed of my child/ward's research/capstone project schedule and off-campus activities, and I hereby give my full consent for their participation.

By signing this waiver, I confirm that I am fully aware of the conditions and potential risks associated with these activities and that I accept full responsibility for my decision.

Parent/Guardian's Name	:	Julie Darlene S. Gromea
Signature	:	
Date	:	March 12, 2026
Contact Number	:	09153823522
Email Address	:	jdarl@gmail.com

Conforme:

Student's Name	:	Nehje John S. Gromea
Signature	:	
Date	:	March 12, 2026

ENDORSEMENT FROM THE UNIVERSITY

Noted:

JAYRELLE B. SY, Ph.D.
IT Capstone Project Adviser
E-mail: *jayrelle.sy@chmsu.edu.ph*

MARITES D. MANGANTI, LPT, Ph.D.
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Data Privacy Consent Form

We are committed to protecting your personal data and ensuring that it is processed in compliance with relevant data protection regulations. All information collected will be used solely for the purposes of this research project and will not be shared with any third parties without your explicit consent. Your participation is entirely voluntary, and you have the right to withdraw from the study at any time.

If you are willing to participate, please indicate your consent by signing and returning the attached consent form to us by ***March 20, 2026***. If you have any questions or require further information, please do not hesitate to contact me at ***09304215678.***

Thank you for considering our request. Your contribution will be instrumental in enhancing the effectiveness of the ***"A.T.L.A.S. (Automated Timetabling and Locations Allocation System): Web-Based Institutional Scheduling Tool."***

Sincerely,

Rowena G. Gilera
Project Manager
Carlos Hilado Memorial State University – Alijis Campus
09304215678 / rowena.gilera@chmsu.edu.ph

Consent Form
I, ***Judy Ann B. Nonato***, hereby grant permission for ***Rowena G. Gilera*** and ***Carlos Hilado Memorial State University – Alijis Campus*** to conduct an interview with me and access relevant data and information related to my use of the ***"A.T.L.A.S. (Automated Timetabling and Locations Allocation System): Web-Based Institutional Scheduling Tool"*** for the purposes of the research project.

I understand that my participation is voluntary and that I can withdraw from the study at any time. I also acknowledge that all data collected will be treated confidentially and in accordance with the data privacy statement provided.

Signature: ____________________
Date: 4/6/2026

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Letter For Beneficiary

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March 16, 2026

JUDY ANN B. NONATO
Principal III
Hinigaran National High School

Dear Ma'am,

Greetings!

We are third-year Information Technology students from Carlos Hilado Memorial State University – Alijis Campus. We are currently working on our IT Capstone Project, which serves as a crucial component of our academic and professional development.

We are writing to formally seek a partnership with Hinigaran National High School to serve as the valued beneficiary of our project. Our goal is to develop and implement a comprehensive, modernized digital infrastructure tailored to your institution's needs.

Specifically, we are proposing a unified digital platform that seamlessly connects four key operational systems:

1. EnrollPro: An Academic Institution's Digital Platform for Optimized Student Admission and Enrollment.
2. A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application.
3. S.M.A.R.T. (Student Management, Academic & Record Tracking)
4. A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation.

By combining these four areas into a single, cohesive platform, we aim to streamline daily workflows, optimize scheduling, and enhance both academic and enhance system process within your institution. Our capstone project is set to run until May 2027. To ensure our system accurately reflects and improves your current workflows, we respectfully request access to relevant operational data and processes, specifically the forms currently used within your existing systems. Please rest assured that all data collected will be handled with the highest level of confidentiality and in strict compliance with our data privacy policy.

Please let us know your available dates and times through this contact details 09304215678, rowena.gilera@chmsu.edu.ph and we will gladly adjust to your convenience. We believe this collaboration will be highly beneficial in ensuring the smooth coordination and successful execution of the project.

Thank you very much for your time, consideration, and support of our academic endeavors. We sincerely hope for your kind approval and look forward to the possibility of working together.

Respectfully yours,

Karyl Ann D. Sudayon	**Rowena G. Gilera**	**Rovin D. Delicano**	**Maika T. Zambra**
Project Leader	Project Leader	Project Leader	Project Leader

Noted by:

JAYRELLE G. SY, Ph.D.
Adviser

MARITES D. MANGANTI, Ph.D., LPT
IT Capstone Instructor/ BSIT Prog. Chairperson

JOE MARIE D. DORMIDO, DIT, PCpE
Dean, College of Computer Studies
Alijis Campus

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College of Computer Studies

BENEFICIARY ACKNOWLEDGMENT AND ACCEPTANCE CERTIFICATE

This is to certify that **Judy Ann B. Nonato,** representing **Hinigaran National High School,** acknowledges and accepts being the official beneficiary of the integrated IT capstone project, which encompasses the following four interconnected systems:

1. EnrollPro: An Academic Institution's Digital Platform for Optimized Student Admission and Enrollment
2. A.T.L.A.S. (Automated Timetabling and Locations Allocation System): A Web-Based Academic Scheduling Application
3. S.M.A.R.T. (Student Management, Academic & Record Tracking)
4. A.I.M.S. (Automated Intervention and Mastery System): A Web-Based Educational Platform with AI-Driven Assessment and Dynamic Remediation

The capstone project, conducted by students from **Carlos Hilado Memorial State University – Alijis Campus** aims to provide innovative solutions aligned with the needs and objectives of the beneficiary. As part of this collaboration, the beneficiary agrees to provide necessary support, feedback, and relevant information to facilitate the successful completion of the project.

By signing this certificate, the beneficiary expresses their consent and commitment to participate in the study and acknowledges the importance of mutual cooperation for the benefit of both parties.

Issued this 16th of March, 2026 at Carlos Hilado Memorial State University – Alijis Campus, Barangay Alijis, Bacolod City, Negros Occidental, 6100.

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JUDY ANN B. NONATO

Signature Over Printed Name/ Date

Designation: PRINCIPAL III Organization: HINIGARAN NHS

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APPENDIX B

Research Instrument

Test Case

A.T.L.A.S. Functional Test Case Evaluation Document

Evaluator Profile:
Name (Optional): ________________________________
Role / Occupation: [ ] IT Consultant / Educator [ ] Systems Analyst [ ] Software Developer
Years of IT Experience: _________________________

General Instructions:
Please execute the following functional test cases within the A.T.L.A.S. system to validate its core technical features. For each scenario, ensure the Preconditions are met, carefully follow the Test Steps, and input the exact Test Data provided. After execution, observe the system's behavior and record it in the Actual Result column. Finally, compare the actual outcome against the Expected Result and evaluate the system's performance using the strict binary scale below:

- **Pass:** The system successfully executed the scenario, and the actual result perfectly matches the expected result.
- **Fail:** The system failed to execute the scenario, required unintended manual intervention, or the actual result deviated from the expected result.

Test Case ID	Test Scenario / Objective	Test Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Status	Remarks
TC-01	Admin Portal (Obj 1.1)	Verify admin login to the web-based administrative portal	Admin account exists	1. Open web portal 2. Enter username & password 3. Click Login	Username: admin Password: password123	User is successfully authenticated and redirected to the Admin Dashboard			
TC-02	Admin Portal (Obj 1.1)	Verify admin login failure with an invalid password	Admin account exists	1. Open web portal 2. Enter username	Username: admin Password: wrongpass	System denies access and displays "Invalid			

				3. Enter wrong password 4. Click Login		credentials"			
TC-03	Priority Management (Obj 1.2)	Verify configuration of a specific scheduling priority	Admin is logged into the portal	1. Navigate to 'Priority Management' 2. Set Max Teaching Hours 3. Click Save	Max Hours: 4	The priority rule is saved successfully to the database			
TC-04	Priority Management (Obj 1.2)	Verify system rejects invalid priority configurations	Admin is logged into the portal	1. Navigate to 'Priority Management' 2. Enter negative capacity 3. Click Save	Room Capacity: -5	System refuses to save and displays an "Invalid Input" error			
TC-05	Teacher Auth (Obj 1.3)	Verify mobile teacher login with valid credentials	Teacher account exists	1. Open mobile web view 2. Enter username & password 3. Click Login	Username: teacher1 Password: pass123	Teacher is successfully authenticated to the mobile dashboard			
TC-06	Preference Submission (Obj 1.3)	Verify submission of teacher schedule preferences	Teacher is authenticated on mobile	1. Open 'Preferences' 2. Select preferred teaching days	Preference: No Afternoon Classes	Preferences are saved and queued for the generation algorithm			

				3. Submit form					
TC-07	Timetable Generation (Obj 1.4)	Verify automated generation of a conflict-free schedule	Priorities and Preferences are saved	1. Navigate to 'Generator' 2. Click 'Generate Timetable'	Complete dataset of teachers, subjects, and rooms	Algorithm outputs a draft schedule with zero double-booked rooms			
TC-08	Timetable Generation (Obj 1.4)	Verify algorithm response to missing critical parameters	Required data is incomplete	1. Navigate to 'Generator' 2. Click 'Generate Timetable'	Missing Data: No teacher assigned to a subject	Algorithm halts generation and displays an error highlighting missing data			
TC-09	Drag-and-Drop (Obj 1.1)	Verify manual schedule refinement	Draft schedule is generated	1. Open visual grid 2. Drag a class block to a vacant time slot 3. Release mouse	Moving Math 7 to a vacant slot	Schedule updates visually, and new time slot is saved without errors			
TC-10	Drag-and-Drop (Obj 1.1)	Verify drag-and-drop clash prevention	Draft schedule is generated	1. Open visual grid 2. Drag a class block into an occupied room slot	Dragging Math 7 to an occupied room	System rejects the move, snaps block back, alerts "Conflict Detected"			

Evaluation Form

RESEARCH INSTRUMENT
A.T.L.A.S. (AUTOMATED TIMETABLING AND LOCATIONS ALLOCATION SYSTEM) SYSTEM EVALUATION SURVEY QUESTIONNAIRE

Type of Respondent: (Please check one)

[ ] School Principal
[ ] Master Teacher / Head Teacher (Scheduler)
[ ] Regular Teacher
[ ] Others: ______________________________

Direction: Below are the statements that will measure the software product quality of a System Project using the ISO/IEC 25010: 2011 standard.

Kindly read the item carefully and please check (/) the appropriate box for each statement that corresponds to the answer you selected using the scale below:

Rating scale:

Code	Interpretation	Description
5	Excellent	4.51 – 5.00. The system is complete and fully functional.
4	Very Good	3.51 – 4.50. The system is complete and functional.
3	Good	2.51 – 3.50 The system is somewhat complete and functional.
2	Poor	1.51 – 2.50. The system is incomplete and somewhat functional.
1	Very Poor	1.50 - below. The system is incomplete and non-functional.

A. Functional Suitability	**5**	**4**	**3**	**2**	**1**
1. **Functional completeness**. Degree to which the set of functions covers all the specified tasks and user objectives.					
2. **Functional correctness**. Degree to which a product or system provides the correct results with the needed degree of precision.					

3. **Functional appropriateness**. Degree to which the functions facilitate the accomplishment of specified tasks and objectives.					
Total Score / Mean Score					

B. Performance Efficiency	**5**	**4**	**3**	**2**	**1**
1. **Time Behavior**. The response and processing times and throughput rates of a product or system, when performing its functions meet requirements.					
2. **Resource Utilization**. The amount and types of resources used by the system, when performing its functions, meet requirements.					
3. **Capacity**. The maximum limits of a product or system parameter meet requirements.					
Total Score / Mean Score					

C**. Usability**	**5**	**4**	**3**	**2**	**1**
1. **Appropriateness recognizability**. Degree to which users can recognize whether a product or system is appropriate for their needs.					
2. **Learnability.** degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
3. **Operability.** Degree to which a product or system has attributes that make it easy to operate and control.					
4. **User error protection.** Degree to which a system protects users against making errors.					
5. **User interface aesthetics.** Degree to which a user interface enables pleasing and satisfying interaction for the user.					
6. **Accessibility**. Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
Total Score / Mean Score					

D. Compatibility	5	4	3	2	1
1. **Co-existence**. Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
2. **Interoperability**. Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.					
Total Score / Mean Score					

E. Reliability	5	4	3	2	1
1. **Maturity**. Degree to which a system, product or component meets needs for reliability under normal operation.					
2. **Availability**. Degree to which a system, product or component is operational and accessible when required for use.					
3. **Fault tolerance**. Degree to which a system, product or component operates as intended despite the presence of hardware or software fault					
4. **Recoverability.** Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.					
Total Score / Mean Score					

F. Security	5	4	3	2	1
1. **Confidentiality**. Degree to which a product or system ensures that data are accessible only to those authorized to have access.					
2. **Integrity**. Degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.					
3. **Non-repudiation**. degree to which actions or events can be proven to have taken place, so that the events or actions cannot be repudiated later.					
4. **Accountability**. Degree to which the actions of an entity can be traced uniquely to the entity.					

5. **Authenticity.** Degree to which the identity of a subject or resource can be proved to be the one claimed.					
Total Score / Mean Score					

G. Maintainability	**5**	**4**	**3**	**2**	**1**
1. **Modularity**. Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.					
2. **Reusability.** Degree to which an asset can be used in more than one system, or in building other assets.					
3. **Analysability.** Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.					
4. **Modifiability**. Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.					
5. **Testability.** Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.					
Total Score / Mean Score					

H. Portability	**5**	**4**	**3**	**2**	**1**
1. **Adaptability**. Degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.					
2. **Installability.** Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.					

3. **Replaceability**. Degree to which a product can replace another specified software product for the same purpose in the same environment.					
Total Score / Mean Score					
Overall Score and Mean Score					